The Effect of the Minimum Wage on Childcare Establishments

by

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Abstract

Childcare is essential for working families, yet it remains increasingly unaffordable and inaccessible for parents and offers poverty-level wages to many employees. While research suggests minimum wage policies may improve the welfare of low-wage workers, there is also evidence they may increase firm exits, especially among smaller, low-profit firms, which could reduce access and harm consumer well-being. This study is the first to examine these trade-offs in the childcare industry, a labor-intensive, highly regulated sector where capital-labor substitution is limited, and to provide evidence on how minimum wage policies affect a dual-sector labor market in the U.S., where self-employed and waged providers serve overlapping markets. Using variation from state-level minimum wage increases between 1995 and 2019 and unique microdata, I implement a cross-state county border discontinuity design to estimate impacts on the stocks, flows, and composition of childcare establishments. I find that while county-level aggregate establishment stocks and employment remained stable, establishment-level turnover increased, and employment decreased. I reconcile these findings by showing that minimum wage increases prompted reallocation, with larger establishments in the waged-sector more likely to enter and less likely to exit, making this one of the first studies to link null aggregate effects to shifts in establishment composition. Finally, I show that minimum wage increases may negatively affect the self-employed sector, resulting in fewer owners with advanced degrees and more with only high school education. These findings suggest that minimum wage policies reshape who provides care in ways that could affect both quality and access.

Keyword: Child Care, Early Childhood Education, Minimum Wage

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1. Introduction

Most minimum wage studies focus on its effects on employment stocks and flows in low-wage industries, finding mixed results for workers overall though positive effects on worker welfare for more affected groups (Neumark and Wascher 2006; Dube, Lester, and Reich 2010, 2016; Cengiz et al. 2019). However, a parallel literature examining establishment-level outcomes finds potential negative effects on consumer well-being, as wage increases often lead to price pass-through to consumers and higher rates of firm closures (Aaronson 2001; Belman and Wolfson 2014; Luca and Luca 2019; Brown and Herbst 2023). In the childcare industry, this trade-off is particularly salient. While wage increases for workers can improve the quality of care, they may also exacerbate disparities among consumers. Families who can afford higher prices may benefit from improved quality, while those with fewer resources face more limited and expensive options. As a result, higher wages may create a divide, producing winners, those who can access higher-quality care, and losers, those who struggle with reduced availability and affordability of care (Malik 2018; Workman 2021; Smith and Osborn 2024). In this highly heterogeneous industry, characterized by strict labor regulations and thin profit margins, minimum wage increases could have different effects than in other low-wage industries, further reducing access to care and potentially affecting maternal labor supply (Anderson and Levine 1999; Morrissey 2017).

In this paper, I examine this concern by providing the first causal estimates of the effect of the minimum wage on childcare establishment stocks and flows in both the waged and self-employed sectors of the industry. I connect 25 years of state minimum wage changes to unique establishment-level data from the Longitudinal Business Database and Integrated Longitudinal

Business Database which contain the near universe of childcare establishments in the United States from 1995 to 2019. I then employ a contiguous county, cross-state border-discontinuity design (Dube, Lester, and Reich 2010, 2016). This model relies on the assumption that contiguous counties on either side of a state policy discontinuity behave similarly prior to a minimum wage increase. By limiting the sample in this way, I ensure the counterfactual counties have similar macro-economic conditions to treated counties thereby overcoming potential spatial heterogeneity that biases traditional county-level two-way fixed effect models.

I begin by examining how an increase in the wage floor affects establishments on both the intensive margin, through changes in employment and payroll, and the extensive margin, through changes in entry and exit rates. Theory suggests that in a competitive market when firms are faced with a wage floor they will respond by substituting labor for capital to offset higher costs, while those unable to make such adjustments will close (Aaronson et al. 2018). Specifically, wage increases directly raise labor expenses, compelling firms to either cut back on hours or staff or pass the costs onto consumers through higher prices. In the childcare industry, these pressures are compounded by strict labor regulations. Childcare establishments must adhere to state mandates, such as minimum child-to-teacher ratios, maximum class sizes, and teacher education requirements, which limit their ability to reduce labor. This leads to labor costs making up 70 percent of a center's expenses, compared to 30 percent in industries like fast food or restaurants (Berebitsky and Rice 2019; Smith and Osborn 2024). As a result, wage increases combined with strict child-teacher ratios and high labor costs could make it more difficult to reduce employment or adjust toward capital, thereby inducing more exits in this industry.

Both intensive and extensive margin effects have significant implications for the quality of care and access for families. Reductions in establishment-level employment could lead to

higher child-to-teacher ratios (if ratios are not currently binding), which are associated with lower quality care and could negatively impact children's short- and long-term outcomes (Phillips et al. 2000; Heckman et al. 2010). Additionally, with 51% of families already living in childcare deserts as of 2019, an increase in firm exits without a corresponding rise in new entries could further restrict access to affordable care (Malik 2018).

I find that a 10 percent increase in the minimum wage leads to a 2.8 percent decline in employment and a 0.29 percentage point increase (or 2.9 percent percent from baseline) in the probability of exit among establishments in the waged sector of the childcare industry. While the negative employment effect is consistent with neoclassical theory, it stands in contrast to many recent studies that find little to no employment response when using aggregate county-level data (Brown & Herbst, 2023; Dube et al., 2010; Ruffini, 2022; Wursten & Reich, 2023; See Appendix A). To assess whether this discrepancy reflects unique features of the childcare sector or simply differences in analytical level, I replicate the analysis using aggregated establishment-level data at the county level, aligning with the approach of prior work. At this level, I find a null effect on overall employment, with an insignificant 1.1 percent decline, which is qualitatively similar to estimates from other low-wage industries, which typically range between -0.7 and 1.1 percent (Brown & Herbst, 2023; Ruffini, 2022; See Appendix A). However, I also find a 4 percent decline in the average number of workers per establishment, suggesting that county-level models may mask dis-employment occurring within firms. These patterns indicate that establishment-level analyses uncover meaningful heterogeneity in adjustment margins that aggregate models obscure. Taken together, the findings suggest that childcare providers may be especially sensitive to rising wage floors, potentially due to the sector’s low margins, high fixed labor costs, and limited ability to raise prices in response.

Researchers attempting to explain the counterintuitive null aggregate employment finding suggest a few possible explanations: low-cost pass-through, elasticity of the labor demand curve, the putty-clay model (i.e., labor to capital shifts), imperfect competitive markets, and aggregate data (Bhasker and To 2003; Manning 2021; Ruffini 2022; Wiltshire, McPherson, and Reich 2023). I show evidence supporting another possible hypothesis, a compositional shift in establishments. This explanation reconciles the null aggregate effect and negative establishment-level effect by suggesting that minimum wage policies may shift the composition of firms, potentially leading to exits among more vulnerable establishments and reallocation of workers to larger or more productive firms within the same industry. Specifically, when the wage floor increases, the overall number of workers in the industry could remain the same even if establishments reduce employment if smaller firms are more likely to exit and larger firms are more likely to enter. This explanation builds on the putty-clay model, which suggests existing firms are less likely to be able to shift from labor to capital expenses which will lead to their closure while new firms will likely be more capital intensive (Aaronson et al. 2018). My hypothesis suggests that while all firms in childcare face higher labor costs with an increased wage floor, smaller firms may be less able to benefit from economies of scale leaving them with less ability to adjust in response to a minimum wage increase than larger firms.

To investigate this hypothesis, I examine whether the minimum wage has differential impacts based on establishment size, chain status, earnings profiles, age, and location. I find that in low-income counties, a 10 percent increase in the minimum wage leads to a 1.1 percentage point increase (or an 11.3 percent increase from baseline) in exits for new and existing smaller, lower-earning, and non-chain establishments, while larger, higher-earning, and chain establishments are unaffected. In high-income counties, I observe a similar pattern of

heterogeneity, but along the intensive margin: the same types of smaller, newer, or lower-earning firms reduce employment, while more established firms do not. These patterns likely reflect the fact that the minimum wage is more binding in low-income areas, where firms face greater constraints in passing on costs. Consistent with this, I find that establishments in low-income counties increase payroll per employee by 3.9 percent following a 10 percent minimum wage hike, while establishments in high-income counties show no significant change. Together, these findings align with other studies suggesting that the minimum wage places greater pressure on smaller firms, pushing them closer to the point where they must either reduce employment or exit the market (Chava, Oettl, and Singh 2019; Luca and Luca 2019; Lee and Park 2023).

Finally, I examine whether establishments in the self-employed sector are affected by minimum wage increases. Unlike other low-wage industries in the United States, the childcare industry includes a substantial self-employed sector that is not bound by minimum wage laws and operates under less regulatory oversight than the waged sector, though some regulation still applies. Economic theory offers competing predictions for how the informal sector might respond to increases in the minimum wage. One view suggests that a higher minimum wage reduces employment in the formal sector, leading displaced workers to enter the informal market. This influx of labor could expand the size of the informal sector and increase earnings for self-employed workers as they compete more directly with formal providers (Welch 1974; Parente 2024). However, alternative models account for "queuing" behavior, which states if rising wages in the formal sector make those jobs more attractive, workers may exit the informal sector in hopes of transitioning to better-paid formal employment, leading to reduced participation in the informal market (Mincer 1976).

I find that while the total number of establishments in the self-employed sector remains unchanged, the composition of the workforce may shift. I find an increase in the probability of workers holding a high school degree or less and a decrease in those holding a bachelor's degree. This change is more pronounced in high-income counties. This finding is interesting, as high-income areas see a stronger effect on employment reduction than low-income areas. Taking into account related literature that finds younger workers with less education are more likely to earn the minimum wage (Neumark and Wascher 2006), these findings could suggest displaced workers from the formal sector are entering the self-employed sector while higher-income workers are potentially leaving the self-employed sector to enter new higher-paying jobs in the formal sector. This finding reconciles the competing economic theories outlined above by adding nuance to how worker mobility between sectors operates differently by a region's income-status.

Related Literature. This paper is related to two literatures, the first examining the effects of childcare regulations on early childcare supply and the second on the effects of the minimum wage on employment stocks and flows. This work builds on research examining the effects of policies impacting firms and workers in the childcare sector, much of which is focused on policies increasing the stringency of staffing regulations (Blau, 2007; Hotz & Xiao, 2011). Brown and Herbst (2023) provide the first estimates of the effects of minimum wage policies on childcare workers using publicly available, aggregate county-level data. In contrast, I use novel establishment-level data, allowing me to explore the impact on establishments rather than workers, with a focus on shifts across counties and variations in establishment characteristics.

I also add to the extensive literature on the minimum wage. While prior studies using county-aggregate data increasingly show minimum wage policies lead to positive effects on

wages but null effects on aggregate employment for more affected groups of workers (Brown & Herbst, 2023; Cengiz et al., 2019; Dube et al., 2010, 2016; Ruffini, 2022; Wursten & Reich, 2023), researchers continue to explore why these findings deviate from economic theory. They are actively working to reconcile the null employment effects by investigating potential explanations, including low-cost pass-through, elasticity of the labor demand curve, imperfect competition in labor markets, and limitations of aggregate data (Bhasker and To 2003; Aaronson et al. 2018; Azar et al. 2019; Manning 2021; Wiltshire, McPherson, and Reich 2023). Unlike more aggregate level analyses, results using establishment-level data are more mixed with some continuing to find null employment effects (Aaronson et al. 2018; Wursten and Reich 2023) while others are able to isolate negative employment effects for more affected firms (Orazem and Mattila 2002; Chava, Oettl, and Singh 2019; Harasztosi and Lindner 2019). There are two key differences in my study. First, I focus on the childcare industry, which has more stringent regulations and a higher labor share, potentially making minimum wage policies more binding in certain regions. Second, I use detailed establishment-level data, allowing me to disaggregate results by both community and establishment-level characteristics, providing more granular insights than studies relying on aggregate data.

Beyond examining negative employment effects, this paper also contributes to the strain of minimum wage studies examining the effect of increased wages on firm viability. While researchers have theorized minimum wage increases would lead to firm exit, research shows the effects highly depend on firm characteristics with wage hikes increasing exit for smaller, lower-quality firms with lower profit margins located in poorer areas (Orazem and Mattila 2002; Rohlin 2011; Aaronson et al. 2018; Mayneris, Poncet, and Zhang 2018; Chava, Oettl, and Singh 2019; Jardim and van Inwegen 2019; Luca and Luca 2019; Dustmann et al. 2021; Wursten and

Reich 2023). Again, my work will examine the extent to which this finding holds in the childcare industry, while addressing additional differences across establishment characteristics including chain status and firm age.

The final strain of the minimum wage literature my paper contributes to is the literature on minimum wage effects in dual economies, building on work in development economics that investigates the effect of regulations in the informal sector. Studies in Brazil and Colombia show wage increases in response to minimum wage policies have different impacts on employment: Brazil experienced a rise in informal jobs, while Colombia saw a decrease (Pérez Pérez 2020; Parente 2024). These differing outcomes may stem from factors like government detection costs and the proportion of the market operating informally making them less likely to generalize to the United States. My work builds on this providing the first evidence within the United States of what happens in an industry with two sectors with different exposure to the minimum wage.

Contribution. This paper makes four key contributions to the minimum wage literature. First, I provide the first causal evidence on the effect of minimum wage increases on individual childcare establishments. While most studies focus on fast food, restaurant, or general low-wage industries, this study uses micro-level establishment data to examine an industry subject to strict staffing regulations, which likely make the minimum wage more binding. For the first time, I disentangle aggregate county-level effects and analyze the characteristics of establishments most affected within the childcare sector. Second, this paper contributes to the literature on minimum wage employment effects by focusing on an industry where arguments that typically rationalize null employment effects, such as labor-to-capital substitution, are weakened due to the industry's stringent staffing regulations. Third, this is the first study to examine minimum wage impacts in

a dual-economy setting within the United States. The waged and self-employed sectors of the childcare industry offer families different qualities of care, making it critical to assess the spillover effects of minimum wage increases on self-employed workers and the potential impact on service quality. Finally, this paper contributes to the small body of literature examining regional differences in minimum wage effects. To my knowledge, only one other study explores differences across low- and high-income counties (Chava, Oettl, and Singh 2019). Understanding regional effects in the childcare industry is especially important, as these could disrupt maternal labor supply in areas already struggling with reduced childcare access.

Layout. The rest of the paper is organized as follows. Section 2 highlights the unique features of the early childcare labor market, focusing on aspects that may lead to differential impacts compared to other low-wage industries, such as staffing regulations and distinctions between the waged and self-employed sectors. In Section 3, I describe the cross-state county-pair border-discontinuity approach and restricted-access establishment-level micro-data used to quantify the effect of the minimum wage on childcare establishments in both the waged and self-employed sectors of the industry. In Section 4, I present the results and in section 5 I conclude.

2. Institutional Context

Early childcare workers care for children between the ages of 0 and 4 and are employed across a diverse range of provider types including: non-licensed (e.g., family members, babysitters), home-based care (single entity, self-employed individuals with no employees)[3], private centers (including those with state funded pre-k slots), Early Head Start and Head Start

[3] In general, home-based care provides care to at least one non-relative for compensation. It is possible for home-based providers to employ an employee. This is less common and is outside of the scope of this project.

(federal funded slots for low-income families), and public school pre-kindergarten or pre-K (state-funded preschool provided by public schools). In this study, I group childcare workers and providers into two sectors: the self-employed or non-employer sector, which consists of all home-based providers, and the waged or employer sector, which includes establishments that employ workers, such as Head Start centers and private centers. In 2019, 59 percent of families with children under five reported using non-parental childcare at least once a week, with many relying on a combination of care options. Among those using childcare, 62 percent used childcare centers at least once a week, 20 percent opted for family day homes, and 38 percent relied on unlicensed relative care (Cui & Natzke, 2021).

In this study, I exclude non-licensed providers and public-school pre-K providers. Although non-licensed providers are an important source of care for some families, the informal nature of the arrangements means it is not captured in data sources with fine grained geography. I am also unable to study public school pre-K programs directly, as public schools typically serve a broader range of grades beyond pre-K, such as Pre-K to 5. This makes it challenging to assess the direct effects of minimum wage changes on pre-K services specifically, as they are integrated within larger institutions serving multiple grades. Public pre-K serves a small, but growing, fraction of all 3-4 year olds (Friedman-Krauss et al. 2018; Cui and Natzke 2021).

Worker Wages and Characteristics

Childcare workers, regardless of sector, earn low wages. The median worker in 2019 earned $13.71 per hour (US Bureau of Labor Statistics 2022). Using the American Community Survey 1-year estimates, I find the state minimum wage is binding for 32% of the industry in

2019,[4] while the average real annual salary for childcare workers has stayed relatively constant since 2000, around $15,000 per year (Figure I).[5] For reference, the federal poverty line in 2015 was $12,490 for a family of one and $16,910 for a family of two (U.S. Department of Health and Human Services 2024). Meaning a single worker earning the federal minimum wage with one child would earn just below the federal poverty line.

Workers in both sectors are primarily white females holding a high school degree or less (Figure II). Since the early 2000s, workers are increasingly likely to be Hispanic and have some advanced education. Racial change is occurring faster in the self-employed sector while educational attainment is growing faster in the waged sector. These changes are likely driven by a combination of broader national trends and the expansion of state and federal policies and regulations affecting the waged sector.

State Regulations and the Impact of Minimum Wage Increases

A critical aspect of the childcare industry is the child-to-teacher ratio, which is more strictly regulated in the waged sector. Both the waged and self-employed sectors must adhere to minimum standards set by state regulatory agencies regarding staffing levels, education requirements, and safety protections (e.g., minimum square footage). However, the self-employed sector faces less rigorous regulations, partly due to enrollment limits that restrict the number of children (or families) served. For instance, while childcare centers may operate

[4] I define the minimum wage as binding for workers whose hourly wage is within 10% of the state minimum wage.
[5] The $15,000 annual salary is based on workers reported salary. If we assume workers work 40 hours per week and 52 weeks of the year, this would translate to a salary of $7.20 per hour. However, the ACS 1-year estimates show the average childcare worker works 30 hours per week, which explains the difference in the median hourly wage reported with the Occupational Employment and Wage Estimates.

multiple classrooms, accommodating 5 to 20 children per teacher per room, family daycare homes are limited by state regulation usually to between six to twelve children.

This difference in capacity makes the child-to-teacher ratio even more relevant in the waged sector. In licensed centers, the required ratios can vary widely. For example, the standard is often one teacher for every five infants, while for four-year-olds, it may rise to one teacher for up to twenty children (US Department of Health and Human Services 2011). In addition to the primary teacher, most states also require centers to employ an additional teacher aide and have a director available for part of the day to ensure compliance with all regulations. Staffing regulations are designed to ensure baseline quality care but inevitably contribute to higher labor costs for waged providers. While the self-employed sector is capped in terms of the total number of children they can serve, they are generally not required to hire aides, and they operate out of their homes, which keeps both labor and capital costs lower. The interplay between child-to-teacher ratios and operational costs underscores the challenges faced by waged providers in maintaining quality care within constrained budgets.

Beyond staffing regulations, the waged and self-employed sectors are differentially impacted by minimum wage policies. The waged sector must adhere to state and federal minimum wage regulations, which means it faces more direct pressure from wage increases. For example, consider a childcare center that serves 70 children with a staff of four teachers, all currently earning the minimum wage. The center maintains a child-to-teacher ratio of 1 to 15 for its two three-year-old rooms and 1 to 20 for its two four-year-old rooms. If the minimum wage increases, the center must decide how to adjust to the new costs. It might choose to maintain its current offerings and pass the additional labor costs onto families, shift its focus to serving only four-year-olds to increase enrollment and stabilize costs, or if the elasticity of demand is high it

might choose to close. Unlike other industries, childcare providers in the waged sector have limited flexibility to shift from labor to capital through technological improvements or reduce worker hours, as they remain bound by regulatory requirements.

In contrast, self-employed family daycare providers, who typically serve a maximum of twelve children, operate outside minimum wage regulations. Their flexible structure allows for adjustments in rates or enrollment with no effect on staffing ratios. However, the self-employed sector could still be impacted; if wages rise in the waged sector, it may incentivize workers to seek higher-paying jobs, potentially drawing them away from self-employment. Conversely, if increased wages lead to reduced employment opportunities in the waged sector, workers might migrate to self-employed roles.

The potential spillover effects on the self-employed sector have important implications for families, especially regarding the quality of care. Families using self-employed providers are more vulnerable to disruptions in coverage due to the lack of substitute staff. Moreover, research underscores the importance of quality care for children's long-term outcomes. Studies like the Perry Preschool Project and evaluations of Head Start programs have shown children who receive high-quality formal care have better long-term outcomes, including improved character skills and academic achievement, compared to children who receive no formal preschool education (Fuller et al. 2004; Heckman et al. 2010; Feller et al. 2016; García et al. 2020).

Current Supply of Childcare Workers, Industry Fragmentation and Market Trends

Beyond childcare worker characteristics and regulations, the current supply of childcare workers is a critical issue. While the 0 to 4 population has declined since 2015, likely due to falling birth rates, childcare employment has steadily increased, apart from the temporary dip

during the COVID-19 pandemic (Figure III). Still, the overall improvement in the ratio of children to childcare workers has been modest, shifting only from 20 to 1 in 2015 to 19 to 1 by 2021. While not all families rely on outside care, these figures highlight a potential strain on childcare supply, which may contribute to stagnation in maternal labor supply, partly attributed to inadequate family-friendly policies and limited childcare availability (Blau & Kahn, 2013).

The childcare industry is highly fragmented, historically dominated by small, single-entity establishments. However, there has been a notable decline in small firms (Figure IV). As of 2021, the nine largest for-profit childcare chains held 10 percent of the market share, expanding their dominance during COVID-19 by acquiring smaller, struggling providers (Haspel 2023). Historically, profit margins in the industry have been around 1 percent, but large firms now report margins of 15 to 20 percent, partly due to strategic pricing in higher-income areas (Grunewald and Davies 2011; Haspel 2023). This shift raises concerns about access and affordability, particularly in low-income regions. If wage increases force smaller centers to close, it could exacerbate the trend of childcare becoming a luxury good in wealthier areas while reducing access for families in lower-income areas.

3. Research Design and Data Sources

3A. Methodology

County Border Cross-State Policy Discontinuity Model

To estimate the effect of the minimum wage on the childcare industry, I exploit cross-state variation in minimum wage policies across all contiguous county-pairs located along state borders using the generalized case study approach popularized by Dube et al. (2010). This model improves upon traditional two-way fixed effect models (TWFE) in two important ways. First, by

restricting the analysis to neighboring counties on opposite sides of state borders, this model better accounts for differential pre-trends by controlling explicitly for county-pair by time fixed effects. Standard TWFE models fail to control for these as they typically include all counties and use only county and time fixed effects. These conventional models overlook time-varying differences across counties, while the contiguous county approach allows for a more precise control of both level differences and time-varying trends between counties. Second, research has shown that counties along state borders tend to be more economically and demographically similar, providing better comparisons than non-border counties (Dube, Lester, and Reich 2010; Allegretto et al. 2013). Minimum wage policies are regionally clustered and affect low-wage labor markets differently. The underlying assumption of parallel trends is strengthened by the fact that contiguous counties face similar local economic conditions, which helps mitigate concerns about confounding macroeconomic shocks or omitted variables and ensure more consistent estimates. By exploiting differences directly across the border between treated and untreated states (or cross-border pairs), I more credibly isolate the effect of wage shifts.

I extend on prior models by estimating the following county- and establishment-level equations (Dube, Lester, and Reich 2010, 2016):

$$y_{cpst} = \beta_0 + \beta_1(lnMW_{s,t}) + \gamma Z_{ct} + \theta_c + \delta_{pt} + \varepsilon_{cpst} \quad (1A)$$

$$y_{ecpst} = \beta_0 + \beta_1(lnMW_{s,t}) + \gamma Z_{ct} + \theta_c + \delta_{pt} + \varepsilon_{ecpst} \quad (1B)$$

Equation (1A) is used to estimate county-level models using all cross-state contiguous county-border pairs. Each observation represents a focal county c in state s (e.g., Steuben County, NY) and its contiguous county in a neighboring state (e.g., Potter County, PA) at time t, where the relationship between the focal and contiguous counties forms county pair, p. Counties can share borders with multiple counties. I include all matched pairs. I analyze a series of county-level

outcomes, y_{cpst}, including total establishments, total employment, employment per establishment, exit and entry rates, first quarter payroll per employment, and total payroll. These measures are derived from establishment-level data (see Section 3B for more detail). All county-level models are weighted by county population. To avoid overweighting counties that appear in multiple border pairs, I allocate the population weight proportionally across pairs (see below for more detail).[6] Equation (1B) is an extension of (1A), where the model is run at the establishment-level. The data is similarly duplicated such that an establishment located in a county that is contiguous to two counties in the bordering state would represent two observations.[7]

$MW_{s,t}$ is the natural log of each state's minimum wage in time t. β_1, the primary coefficient of interest, represents the effect of a 1 percent increase in the minimum wage on the outcome of interest. I scale all reported effects by a factor of 10 to reflect the impact of a 10 percent increase in the minimum wage, a more realistic policy change. Each model includes county, θ_c, and county-pair time, δ_{pt}, fixed effects. Controls for time-varying county controls, Z_{ct}, including logged population of 0-4, 5-19, and 20-64-year olds, pre-K enrollment, and private sector employment.[8,9] I use two-way clustered standard errors, clustering on both state and state

[6] I run sensitivity tests using standard population weights and find the results are not overly sensitive to this adjustment (see Appendix D).

[7] I do not include establishment fixed effects. While the introduction of establishment fixed effects would help to absorb all time-invariant characteristics of establishments and isolate the within-establishment variation over time, these models would exclude establishments that are only in the sample for 1 year which represent an important part of the market. Results from models including establishment fixed effects vary for select outcomes and are available upon request.

[8] Throughout this paper, all log transformations are the natural log. Because some counties have zero pre-K enrollment over my study period and taking the naïve log would force these observations to become missing, I transform pre-K using the natural log plus one. Private sector employment is calculated using county total employment derived from the Longitudinal Business Database.

[9] Following prior minimum wage studies, I conduct several sensitivity tests to assess the robustness of the results. These include: (1) estimating models without pre-K enrollment, to balance the concern that public pre-K may crowd out private childcare (Bassok et al. 2014) but might also be endogenous; (2) restricting the sample to county pairs with centroids within 75 kilometers to better capture local labor market dynamics; and (3) including controls for recessionary conditions, which may independently influence childcare outcomes. The main findings are not robust to these alternative specifications (see Appendix D for details).

border segment. Clustering at this level allows me to account for correlations in the residuals of both state and state-border segments. For instance, there could be unobserved factors that similarly affect counties within a single state (or counties along a specific state border), and clustering accounts for these potential correlations to ensure estimates are not biased downward.

While this model helps ensure the parallel trends assumption holds, there are three other assumptions that must hold to ensure consistent estimates. First, there should be no contemporaneous policy shocks. Over the study period (1995–2019), there were several notable external shocks that may have impacted the childcare industry. First was the economic recession of 2007-2009. The childcare sector is particularly sensitive to economic downturns, which can affect demand for services and employment in the sector (Brown & Herbst, 2022). Second, federal funding for the Child Care and Development Fund (CCDF) gradually increased over time and was distributed to states using a stable formula based in part on the number of children under age 13. While some variation in funding across counties in different states could occur due to demographic differences, I include county-pair-by-year fixed effects, which absorb time-varying state-level policy differences. Additionally, I control for the population of 0 to 4-year-olds, which captures much of the relevant variation in young child population likely to influence center-based care and funding needs. Together, these controls help limit concerns that differential state funding trends drive the observed results. Outside of these policies, no states to my knowledge adopted additional policies to help fund childcare or change the relative demand and supply of care.[10]

[10] The timeframe of this study stops in 2019 specifically to avoid the differential impact of COVID-19 and the resulting American Rescue Plan funding as well as state initiatives such as DC's Pay Equity Fund and New Mexico making childcare free for all.

Second, I explore the extent to which there is treatment spillover and the potential violation of the stable unit treatment value assumption (SUTVA). This assumption requires the treatment assigned to one unit does not affect other units. In this context, the possibility of spillover effects arises if firms or workers relocate to avoid the effects of minimum wage increases. For instance, if a minimum wage increase occurs in New York but not Pennsylvania, businesses might consider moving to Pennsylvania to avoid paying higher wages or workers might try to enter New York to receive higher wages. Either movement would violate this assumption. Unlike a fully tradable sector, childcare providers face a series of fixed costs including state licensure and inspections that would make a move across state borders difficult and costly. Childcare is also highly localized, with most families driving less than 5 miles from their home to care (National Survey of Early Care and Education 2016). This suggests the likelihood of spillover, in terms of firms or families moving across state border, is relatively low.

The final assumption requires there is no anticipation effect, meaning treated states should not alter their behavior before the minimum wage changes due to anticipation. In the first seminal minimum wage study using a difference-in-differences approach, Card and Krueger (1994) point to the fact that the minimum wage change in New Jersey was completely unexpected to assuage concerns that businesses or workers anticipated the change and reacted prior to the policy. However, a recent trend among states is the introduction of minimum wage schedules, where wage increases are phased in over time. Consider, California who introduced a bill in 2017 that legislated annual increases until the state achieved a $15 minimum wage after which the minimum wage would be tied to inflation. A handful of other cities and states including Washington, DC in 2015, New York, NY in 2014, and Massachusetts in 2014 passed similar policies. These schedules make policy changes more predictable, increasing the

likelihood that employers and workers adjust behavior in advance—such as altering hiring, pricing, or investment decisions before the wage officially rises. Such anticipation could bias estimates toward zero by shifting part of the policy response into the pre-treatment period, thereby attenuating the estimated effect. While the inclusion of county-pair-by-time fixed effects helps mitigate this concern to some extent, it cannot fully address the bias introduced by anticipatory behavior.[11]

I also estimate event study and stacked event study models to assess the sensitivity of my estimates to potential biases highlighted in recent difference-in-differences literature. The results are highly similar, suggesting minimal bias from treatment effect heterogeneity or staggered adoption. Therefore, I present estimates from the standard difference-in-differences specification. See Appendix B for details.

I next expand the previous establishment-level models to explore heterogeneous treatment effects across establishments in the waged sector. Equation (2A) is an update to equation (1B), where I interact the logged minimum wage with establishment-level characteristics, Est_Char_{et-1}.

$$y_{ecpst} = \beta_0 + \beta_1(lnMW_{s,t}) + \beta_2(Est_Char_{et-1}) + \beta_3(lnMW_{s,t}x\ Est_Char_{et-1}) + \theta_c + \delta_{pt} + \partial_e + \varepsilon_{cpst} \quad (2A)$$

I focus on four primary establishment-level characteristics, an establishment's earning's level, employment size, age, and chain-status. For each characteristic, I construct a categorical variable, Est_Char_{et-1}, with three mutually exclusive categories. The first category is always composed of establishments who are new in year t. The second and third categories include

[11] I stop the current analysis in 2019 to avoid COVID-19 as well as a slew of additional states who passed legislation around that time introducing minimum wage schedules including Delaware, Florida, Illinois, Maryland, New Jersey, New York state, and Rhode Island.

establishments that were active in year t-1 whose lagged characteristics are above or below a predetermined threshold.[12] The coefficients of interest are β_1 and β_3. β_1 can be interpreted as the baseline effect of the minimum wage for new establishments while the sum of β_1 and β_3 is the net effect of the minimum wage on existing establishments with a given lagged characteristic.

For earnings, the threshold is whether the establishment's lagged average earnings per worker fall within 10% of the state minimum wage. This serves as a proxy for whether the establishment is likely to be directly affected by a minimum wage policy. While informative, this proxy is imperfect because I do not observe hours worked. Establishments that rely heavily on part-time workers may appear to pay low average earnings per worker, even if their hourly wages exceed the minimum wage. As a result, the measure may overstate their exposure to the policy.

For employment size, I classify establishments based on whether they had fewer than 20 employees or 20 or more in the prior year. Larger establishments may be better positioned to withstand minimum wage increases for several reasons. They often hold greater market share or possess more pricing power, enabling them to pass through increased labor costs to consumers. They may also have more diversified operations or revenue streams, which can buffer against cost shocks. Operationally, larger firms are potentially more likely to adjust on the intensive margin by reducing employment, while continuing to operate, whereas smaller firms may lack the staffing flexibility to make such adjustments pushing them instead to exit the market. Additionally, larger firms may benefit from economies of scale, such as more efficient scheduling and human resources practices, which can mitigate the per-worker impact of wage increases.

[12] For each establishment characteristic, I also estimate models using a data-driven threshold, classifying establishments as above or below the median based on their lagged values, rather than relying on a predetermined cutoff. Results are generally similar and available upon request.

A similar rationale applies to an establishment's chain status. Chain-affiliated establishments may benefit from centralized management and standardized operations, all of which can reduce the unit cost of adjustment and increase resilience to policy shocks. Chains may also have greater financial resources or cross-subsidization across locations, allowing them to sustain temporary losses more easily than independent establishments.

For establishment age, I define "young" establishments as those that have been operating for 1 to 3 years and "established" ones as those operating for 4 or more years. This threshold balances power while aligning with empirical evidence that newer establishments face a higher risk of failure. Young firms often contend with uncertain demand, limited customer loyalty, and underdeveloped internal processes. They may also have less access to credit or capital and face steeper learning curves in managing labor and regulatory costs. As a result, younger establishments may be less able to adjust or absorb wage shocks, increasing their vulnerability to exit following minimum wage increases.

3B. Data

The Longitudinal Business Database and Integrated Longitudinal Business Database

Recent minimum wage studies examining the effect of policy changes on business dynamics within the United States have relied primarily on three sources of data: national data from the Quarterly Census of Employment and Wages (QCEW;(Aaronson et al. 2018)) or Quarterly Workforce Indicators (QWI; (Wursten and Reich 2023)); private data from Dun and Bradstreet (Rohlin 2011; Chava, Oettl, and Singh 2019); and rich state administrative payroll data (Orazem and Mattila 2002; Jardim and van Inwegen 2019). Each data source measures establishment

employment by industry with varying levels of specificity on employment by worker demographics.

While each of these provides detailed information on employers, none of these sources include self-employed workers. The self-employed workforce composes 10% of all workers, and within childcare this number is likely even higher (Hipple and Hammond 2016). The National Center for Education Statistics (NCES) estimates roughly 20% of children receive nonparental care from family day home providers, or providers who are self-employed, at least one day a week (Cui and Natzke 2021).

To bring information about this important sector to bear, I combine the Longitudinal Business Database (LBD) and Integrated Longitudinal Business Databases (ILBD) to examine, for the first time in the childcare setting, the effect of minimum wage policies on the non-employer market. The LBD and ILBD include well-validated establishment identifiers that track establishments as they move from employer to non-employer status. This allows me to track establishments across sectors, to control for a rich set of characteristics, and examine differences in outcomes with a more accurate measure of the tenure of establishments.

I use micro-level Census and IRS records spanning 1995 to 2019 including all employer and non-employer businesses from the Longitudinal Business Database (LBD) and Integrated Longitudinal Business Database (ILBD), respectively.[13] I combine these sources to create a longitudinal yearly establishment-level panel from 1995 to 2019 containing establishment, or provider, entry and exit indicators for the entire childcare industry.[14] Both the LBD and ILBD

[13] The Longitudinal Business Database was redesigned in 2002 causing an interruption in some of the business identifiers. I use the Census' researcher-derived uniform business identifier that spans this time range.
[14] All data from the Longitudinal Business Database with a NAICS, or industry, code of 624410 is considered a childcare center. All data from the Integrated Longitudinal Business Database with an industry code of 624410 is classified as a family day home. Prior to 1998, the ILBD used the Standard Industrial Classification (SIC) instead of the NAICS. For this period, I define all childcare services as those with a SIC of 835, or Child Day Care Services.

report employment annually, so my analysis is at the annual level. Detailed information on the LBD and ILBD and how the respective panels were created is in appendix C. I begin the analysis in 1995 because, as noted in Cengiz et al. 2019, most state minimum wage changes begin in the 1990s. The sample stops in 2019 because of the many COVID-19 related shocks to the childcare industry, including significantly altered demand during shutdowns and a historically large stimulus funding package distributed heterogeneously across states.

Using these data, I construct two analytic datasets: (1) establishment-level panels for the waged and self-employed sectors, and (2) county-level panels created by aggregating the establishment-level data to align with publicly available statistics. Below, I describe the outcome measures constructed from each dataset.

County-Level Outcomes

I examine trends in establishment dynamics using the following dependent variables: (i) Total establishments: The total number of establishments in county c in year t; (ii) Total employment: The total number of workers employed in county c as of March 12th of year t;[15] (iii) Establishment-level employment: The median number of workers employed at all establishments e in county c as of March 12th of year t; (iv) Median first quarter payroll divided by employment: For the employer sector this is calculated as the median across every establishment's payroll in the first quarter (in 2021 USD) divided by the total employment in establishment e for county c in year t weighted by the establishment-level employment.[16] For the

[15] This measure is only valid for the LBD employer sector, as each record in the ILBD counts as single-employee entity.

[16] In the LBD payroll is reported as a total across all employees per $1000. I transform this by taking total payroll times $1000 divided by total number of employees. Both average payroll and income are transformed to 2021 real dollars, or present value using the formula: $Salary_t/CPI_t * CPI_{2021}$).

non-employer sector this is calculated as the median total income net receipts (in 2021 USD); (v) Median payroll: The median of the total payroll across all firms weighted by the total number of establishment-level employment.[17] (vi) Firm exit rate: The total number of establishments e in county c present in year t with non-zero employment and positive payroll who were not present in t+1 divided by the total number of establishments in county c with positive employment and payroll in year t-1.[18]; and (vii) Firm entry rate: The total number of establishments in county c that are not present in t-1 but present in time t with non-zero employment and positive payroll divided by the total number of establishments in t-1 in county c.

Each of these measures is then transformed as follows. Counties with no employment, establishments, or payroll all are given a value of zero in the respective year they have no positive value and are transformed using the inverse hyperbolic sine so they contribute to the model estimation.[19] Establishment-level employment and first quarter income are transformed using the natural log and counties with no employees are excluded from the analysis. The exit and entry rates are proportions; therefore, they do not undergo further transformation.

Establishment Outcomes

I complement the county-level analysis with an establishment-level analysis examining trends in firm dynamics using the following dependent variables: (i) Exit probability: Defined as an establishment that is present in year t with non-zero employment and positive payroll but not present in t+1; (ii) Entry probability: Defined as an establishment that is not present in t-1 but present in time t with non-zero employment and positive payroll; (iii) Employment; Defined as

[17] This is only valid for the LBD. The ILBD only reports income since they are self-employed entities.
[18] Using *t-1* as the denominator avoids conditioning on the potentially policy-affected firm population in year *t.*
[19] Due to recent concerns about the interpretation of the inverse hyperbolic sine, I also log transform these variables and find no meaningful difference in the magnitude or direction of effects (Chen and Roth 2024).

the total number of workers at establishment e as of March 12th; (iv) Average employee earnings: For the employer sector this is calculated as the establishment's payroll in the first quarter (in 2021 USD) divided by the total employment in establishment e in year t. For the non-employer sector this is calculated as the reported income net receipts (in 2021 USD)[20]; (v) Total payroll: This is calculated only for the employer-sector and is the total establishment payroll (in 2021 USD). Both employment and income are transformed using the natural log for the waged sector. Income is transformed using the inverse hyperbolic sine for the self-employed sector, since employers could report an income of zero.

Border Pair

The primary research design in this paper relies on contiguous border county pairs with cross state policy variation. The LBD and ILBD each represent every tax-filing establishment in the United States, and thus cover every county. There are 3,143 counties in the United States, and of these 1,184 sit on a state border (Appendix D). Some counties border multiple counties resulting in a total of 1,306 unique county pairs.[21] To create the border sample, I use the United State Census' 2010 County Adjacency File (US Census Bureau 2010).[22] This file contains a list of every county pair in the United States, including counties that are adjacent within the same state and across state lines. I take this dataset and limit it to only counties across state lines. For each county-pair, I calculate both the distance in kilometers between the centroids of both

[20] For the ILBD self-employed establishments, I exclude 0.5% of establishments with extreme payroll defined as non-positive or greater than $200,000.
[21] Roughly 50% of counties are adjacent to two counties with roughly 20% each adjacent to one and three counties. The maximum number of adjacent counties is 9, though counties with 6 or more adjacencies represent only 1% of all counties.
[22] County adjacency file 2010 used: The analysis uses data from 1994-2020. Each state has their own policy for approving county border adjustments. Over the past 30 years the number of substantial county border changes (or changes affecting 200 or more people) has been relatively small. See a full list by the Census here https://www.icip.iastate.edu/tables/population/states-estimates

counties as well as the distance of the intersection at the border. For instance, if county A in state A touches county A in state B, county B in state B, and county A in state C, then I would calculate the intersection between county A in state A and each of these three contiguous counties. I then convert the intersecting length into a percent of county A in state A's total contiguous state border. I use this file to limit the yearly provider-level panel to only border counties. Using this connected dataset, I then run analyses using all border county pairs. Following previous studies, I test for balance among the sample of contiguous county pairs compared to all counties on a series of county characteristics using a simple difference-in-means estimation (Table I). I find the border county's characteristics are similar to those of all counties in terms of their relative population characteristics and childcare industry characteristics. This suggests the findings from this study may generalize to non-border counties.

Minimum Wage Policy

To examine how changes in compensation affect establishment dynamics, I exploit annual state-level variation in the minimum wage for all events regardless of size (Figure V). To do this, I create a yearly state minimum wage panel using publicly available data (Vaghul and Zipperer 2016).[23] The minimum wage is defined as a state's average monthly minimum wage for the first quarter or across all months in the year. The minimum wage is then logged. The first quarter logged minimum wage is used as the independent variable for outcomes defined as of the first quarter (or as of March 12th), whereas the yearly logged minimum wage is used for all other outcomes.

[23] Beginning in 2004, cities began instituting their own minimum wage policies. Future work will consider the effect of substate minimum wage policies. Substate minimum wages can be defined at varying geographies depending on the state (e.g., city, county, non-urban or urban regions). Because many substate minimum wage policies do not occur on state borders, this set of analyses will require a slightly different research design.

Weights

The border discontinuity model requires each county to be repeated N times where N is the number of unique contiguous county pairs. Recent county-level minimum wage studies that have incorporated border discontinuities have applied raw county population weights, however these weights could overrepresent counties with multiple county-pairs (Brown & Herbst, 2023). To avoid this, I weight for county-population by the percent of a county pair's shared length. Specifically, for each contiguous county border pair, I calculate the shared distance between the two counties. Then for each focal county, I calculate the total shared distance across all county pairs. I then determine the percent overlap for each contiguous pair by taking the pair's distance and dividing this by the focal county's total shared distance. I then multiply this share by the total population weight, so that the total population is down weighted. For establishment-level models, I only weight by the share overlap. I do not multiply this by the total population, as the frequency of establishments in the establishment-level panel should proxy for the county's population. To show how weighting affects the results, I run robustness tests with traditional population weights (Appendix E).

Controls

Most minimum wage studies control for county's total private employment and population to alleviate concerns about broader trends that could contribute to changes in employment that are not picked up by pair-time and county fixed effects. In examining the childcare industry there are a few additional concerns including how changing natality may affect demand for childcare. Additionally, there is evidence the childcare industry is more sensitive to the business cycle than

other low-wage industries and increasing access to universal pre-K could crowd out private childcare (Bassok et al., 2014; Brown, 2018; Brown & Herbst, 2022).

To address these concerns, I include the following controls in my models: (i) yearly county population of 0 to 4-year-olds, 5 to 19-year-olds, and 20 to 64-year-olds. To create these controls I combine data from the US Intercensal County Population Data from 1994 to 2016 and the American Community Survey 5-year estimates from 2017 to 2022 (Roth 2016; US Census Bureau 2021). I choose these groups to represent the population of childcare aged children, school-aged children, and working aged individuals. I take the natural log plus one of each population count; (ii) yearly county private employment. To create this measure, I aggregate the total employment across all establishments in the LBD and then take the natural log; and (iii) yearly county pre-K enrollment. I create this measure using the Common Core of Data and I transform it using natural log plus one (US Department of Education 2021). I include this control because public pre-K is growing over the timeframe of this study. While it could be the case that pre-K is growing in response to the minimum wage hikes making this a bad control, most expansions in the pre-K sector are not correlated with minimum wage policies. Moreover, public pre-K teachers are typically paid on the same salary schedule as other K-12 public school teachers, though their actual earnings may be lower due to differences in education, experience, and hours worked. In 2023, pre-K teachers earned an average salary of $41,410, higher than the $24,040 earned by childcare workers but lower than the overall K-12 teacher average of $74,410 (US Bureau of Labor Statistics 2024). This pay disparity suggests pre-K teachers might be unaffected by minimum wage changes. I run robustness checks without pre-K enrollment and show its inclusion does not make a substantive difference to the results (see Appendix E).

Researchers also explore various ways of controlling for the Great Recession of 2007 to 2009 by omitting these years from their models, interacting state fixed effects with individual year dummies for the three affected periods, or controlling for the intensity of the Recession by including the Federal Housing Finance Agency's Housing Price Index (Dube, Lester, and Reich 2010, 2016; Clemens and Wither 2019). To alleviate concerns that the effects I run sensitivity checks including state recessionary dummies (see Appendix E).

Heterogeneity

Childcare providers are likely to experience differential effects based on their own establishment-level characteristics or the market in which they operate. Therefore, I examine differences in outcomes for establishments across multiple dimensions. First, I examine differences by childcare sector: family day homes and childcare centers. Childcare sectors are likely differentially affected by the minimum wage. Family day homes operating with no employees are not obligated to increase their wages when new minimum wages are enacted. Depending on the level of competition in the market, one could theorize an increase in the minimum wage could result in either no change or a slight increase in the unregulated, family day home market. To measure the effect of the minimum wage across sectors, I create indicators for non-employer family day homes and employer childcare centers. I define all childcare sectors as those establishments in the LBD with NAICS 624410 and all establishments with an industry code of NAICS 624410 in the ILBD who are sole proprietors with no matching record in the LBD as family day homes.[24]

[24] NAICS 611110 does not appear in the ILBD.

Second, elasticity of demand due to parent willingness or ability to pay for childcare could vary by location, which could greatly affect the providers' ability to adjust tuition following a minimum wage increase. To unpack this, I estimate the effect for establishments in low-income versus high-income regions as defined by whether a county's 2016 household median income is above or below median as reported in the US Census' American Community Survey.

Third, there are likely establishment characteristics which could lead to differential effects of the minimum wage. For the waged sector, I examine whether establishments that are younger, paying closer to the minimum wage, are non-chains, or employ fewer workers could be more at risk of closing in response to minimum wage increases. To determine the extent to which these features impact establishment's behavior, I create a categorical variable where each level represents whether an establishment is new to the sample or whether the establishment has been in the sample for two or more years and is either below or above the pre-determined lagged cut-off (see section 3A for more detail).

Finally, for self-employed establishments, I turn to the demographics of the owners themselves to see whether minimum wage changes shift who is leaving or entering this workforce. The Longitudinal Employee Household Database collects information on individuals' education levels from a variety of sources such as Census Surveys. I link these records to the owners of the self-employed childcare establishments and create indicators for whether an owner holds a high school degree or less, some college, or a bachelor's degree or more.[25]

4. Results

[25] For education, I limit to only observed (non-imputed) values. I limit to only workers who are 25 years or older per the Census' recommendation, which is based on the age where education-status becomes increasingly stable.

Minimum Wage Impacts on Employment, Wages, and Firm Dynamics in the Waged Sector

In a competitive labor market, neoclassical theory predicts that increasing the minimum wage should raise wages, reduce employment, and increase exits among labor-intensive firms. I assess these predictions in the context of the childcare industry by analyzing effects on employment, earnings, and establishment dynamics including entry and exit. A central contribution of this paper is to bridge existing work, which relies almost exclusively on aggregate county-level data, with newly available establishment-level administrative data. This comparison allows me to test whether standard county-level estimates mask within-county heterogeneity or adjustments that establishments make in response to minimum wage increases.

Consistent with contemporary studies, I find the minimum wage does not lead to a decline in the total number of childcare establishments or total employed in the waged sector. For context, the average county has about eight establishments in the waged childcare sector, with roughly 10 percent of firms entering or exiting each year (Table II). These establishments employ an average of 12.1 workers earning $3,641 per quarter, or approximately $14,564 annually. County-level results show no statistically significant effect of the minimum wage on total establishments or employment (Table II, Columns 1-4). However, the point estimate is slightly negative for employment with a 10 percent increase in the minimum wage associated with a 1 percent decline in total employment. While not statistically distinguishable from zero, this estimate is in the range of recent findings from contemporary minimum wage studies, which report null employment effects ranging from -0.7% to 1.1% (see Appendix Figure A.1).

One limitation of county-level estimates is that they reflect net changes in total employment across all establishments and may obscure heterogeneity in firm responses. For instance, large providers might expand while smaller providers reduce employment or exit

entirely. To explore this possibility, I examine the impact of the minimum wage on median employment per establishment. At the county level, I find a 10 percent increase in the minimum wage reduces the average number of workers per establishment by 4 percent (Table II, Column 5). This effect is statistically significant at the 10 percent level. Taken together with the null effect on total employment, this pattern is suggestive of a shift in the distribution of firm sizes: smaller firms may be contracting or exiting while larger firms expand, leading to reallocation that offsets average declines. I explore this hypothesis more in the following sections.

To more directly test whether these patterns hold at the micro level, I estimate analogous models using establishment-level administrative data. These models are not weighted by establishment size, meaning the estimates are average treatment effects. I find a 10 percent increase in the minimum wage leads to a 2.7 percent decline in employment, somewhat smaller than the county-level 4 percent reduction in median employment (Table III, Column 1). This is a reduction of roughly 0.2 workers relative to baseline, suggesting that for every five establishments, an average of 1 employee will lose their job. Overall payroll falls at a commensurate rate with employment (Table III, Column 5).

Turning to wages, I proxy for earnings using first-quarter payroll per worker. While many studies report increases in average wages following a minimum wage hike, I find much smaller effects in the waged sector of children (see Appendix A). Based on American Community Survey data, I estimate that about 32 percent of childcare workers earn within 10 percent of the minimum wage. If these workers receive a full 10 percent increase in wages, and there are no spillovers or job losses, the average wage would rise by up to 3.2 percent. However, the actual change in average wages depends on the broader wage distribution, any spillover effects, and potential dis-employment, so this figure should be interpreted as a rough upper bound. However,

I find only a 0.39 percent increase in county-level earnings (Table II, Column 6), and this estimate is statistically insignificant. The establishment-level models show a similarly null effect (Table III, Column 2).

There are several possible explanations for this muted wage response. One is that firms comply with higher minimum wages by cutting hours, thereby keeping payroll per worker constant. Another is that firms offset wage increases by reducing non-wage compensation (e.g., benefits which are exceedingly rare in childcare). It is also possible that minimum wage increases lead to selective job loss among the lowest-paid workers. Finally, these average effects may mask variation across counties or firms, a possibility I explore in the next section.

Finally, I examine the effect of the minimum wage on firm dynamics, specifically entry and exit. At the county-level, I find no change in the total number of firms, but I do observe increased churn. A 10 percent increase in the minimum wage is associated with a 0.37 percentage point increase in the county exit rate (Table II, Column 7), nearly a 4 percent increase relative to the baseline exit rate of 10 percent. Although the entry rate estimate is of similar magnitude, it is not statistically significant. These results are consistent with neoclassical predictions and suggest that some firms are unable to absorb higher labor costs and exit the market. The establishment-level model yields similar results, though slightly attenuated: the exit rate rises by 0.27 percentage points (Table II, Column 3). In 2019, there were roughly 77,000 waged childcare centers operating (Figure III). This means that in a typical year 7,700 will close and a 10% increase in the minimum wage will lead to an additional 223 closures.

Taken together, these findings highlight the importance of combining county- and establishment-level perspectives. County-level models are useful for benchmarking against the existing literature, but they can obscure adjustment margins within counties. Establishment-level

data allow for a more granular view of how firms respond, especially on the intensive and extensive margins. The next section explores which types of establishments and regions drive the observed aggregate effects.

Minimum Wage Impacts on the Waged Sector Across Low- and High-Income Counties

Childcare is increasingly expensive with many families struggling to afford high-quality care (Workman 2021). A key factor that may shape how childcare providers respond to minimum wage increases is the elasticity of demand for their services. In areas with more inelastic demand, typically higher-income communities, providers may be able to pass increased labor costs on to families without substantial loss in enrollment. In contrast, in lower-income areas with more elastic demand, rising labor costs may be harder to absorb, potentially leading to establishment closures. To examine this, I stratify the sample by county income level, using the national median income to classify counties as either low-income or high-income. This division serves as a proxy for differences in local demand elasticity and capacity to absorb cost shocks.

The structure of the childcare market varies substantially across low- and high-income counties. There are an average of 15 establishments employing 134 workers in high-income counties, compared to 5 establishments and 33 workers in low-income counties (Table IV). Importantly, high-income counties have roughly 4 times as many children aged 0-5, which suggests overall capacity is roughly similar across county income-status. Wages are also higher. Workers in high-income counties earn $382 more per quarter, on average. However, both groups of counties share some important similarities. Entry and exit rates are roughly equal (around 10 percent), and the typical establishment remains small, averaging about 12 employees.

Despite these similarities, the response to minimum wage increases diverges across county income levels. In low-income counties, a 10 percent increase in the minimum wage is associated with a significant 4 percent increase in first-quarter payroll per worker, while earnings in high-income counties remain unaffected (Table IV, Column 2).[26] One possible explanation is that a larger share of workers in low-income counties may be paid at or near the minimum wage, making these establishments more directly affected by minimum wage policies.

There are also meaningful differences in how counties adjust along the intensive and extensive margins. Establishment-level estimates suggest that minimum wage increases lead to a statistically significant 1 percentage point increase in the exit rate in low-income counties, while employment declines by an insignificant 1.9 percent (Table IV, Columns 1 and 3). In contrast, high-income counties experience a smaller, though still statistically significant, increase in exit rates (0.24 percentage points) and a larger, significant decline in employment of 3 percent.

Although the confidence intervals on the employment and exit effects overlap across income groups, meaning I cannot reject the null hypothesis that these effects are statistically different, qualitatively the patterns are consistent with the theory that low-income counties, where the minimum wage is more binding and demand more elastic, are more likely to adjust on the extensive margin by exiting. In contrast, high-income counties are more likely to adjust on the intensive margin, reducing employment per establishment. This finding supports the broader theoretical claim that the local demand environment shapes how childcare providers absorb labor cost shocks.

Minimum Wage Impacts by Establishment Characteristics

[26] Findings are similar when using county-level aggregate data (see Appendix F).

Economic theory predicts that establishments more bound by the minimum wage, either because they employ more minimum wage workers or because labor represents a larger share of total costs, should be more affected by wage mandates. To proxy for these dimensions of vulnerability, I examine heterogeneity in treatment effects by establishment's lagged first-quarter payroll per employment (a proxy for average earnings), employment size, chain status, and firm age. I examine the extent to which these dimensions affect exit probability, employment, earnings, and entry in turn.

Exit Probability

I find firm earnings, size, chain-status, and age play an important role in moderating exit risk, particularly in low-income counties. In these counties, a 10 percent increase in the minimum wage increases the exit rate for new firms by about 1.2 percentage points (Table V-VIII, Column 1). I find the point estimates for existing lower-earning, smaller, and non-chain establishments are not statistically different from those of the new firms. Existing higher-earning, larger, older, and chain establishments, however, are differentially affected. Along each establishment characteristic, the effect on exit probability is significantly lower than that of new firms, with reductions ranging from 27 to 62 percent relative to new firms. This suggests that minimum wage increases may crowd out less competitive firms while higher-earning establishments remain resilient or even benefit.

In contrast, high-income counties show no significant relationship between minimum wage increases and exit probabilities (Table V-VIII, Column 5). This may reflect greater demand in these areas, a higher ability to pass on costs to consumers, or the fact that they might be more likely to adjust on the intensive margin or employment.

Employment

Turning to employment, high-income counties exhibit stronger and more precise dis-employment effects. A 10 percent increase in the minimum wage leads to a significant 2.6 percent decline in employment among both new firms and existing firms with lower lagged earnings (Table V, Column 6). By contrast, the coefficient for existing higher-earning firms fully offsets the negative effect observed for new firms, implying no measurable employment loss for these establishments. While the estimated increase in employment (1.4%) is not statistically significant on its own, it is significantly different from the decline observed among new firms, indicating meaningful heterogeneity in how firms respond to minimum wage increases.

This pattern extends across other firm characteristics. Newer, smaller, and younger firms exhibit point estimates that are not statistically different from the baseline group of new firms, suggesting they, too, respond to minimum wage increases by reducing employment. In contrast, larger, older, and more established firms show significantly smaller dis-employment effects relative to new firms, implying they are less constrained in their employment decisions following minimum wage increases.

These results suggest that firms with fewer resources or less market stability, whether measured by age, size, or earnings, may be more constrained by rising labor costs, while better-resourced firms are not. In low-income counties, a similar pattern emerges, though the estimates are noisier and less precisely estimated (Tables V-VIII, Column 2).

Earnings Proxy

Results for first-quarter payroll per employee, a proxy for earnings, generally align with the patterns observed for exit probability, though some deviations emerge. In low-income counties, all establishment types experience increases in average earnings following a minimum wage hike. New firms see a 3 percent increase, while existing firms with lower and higher lagged earnings see increases of 5 to 6.5 percent, respectively (Table V, Column 3). These patterns are consistent across firm size, age, and chain status, though the effect for new firms is less precisely estimated and not statistically significant in all specifications. These findings suggest that in low-income counties, establishments across all characteristics respond to minimum wage increases by raising earnings. This is consistent with the idea that most are operating near the minimum wage, leaving little room for differential responses.

In high-income counties, by contrast, the pattern diverges. A 10 percent increase in the minimum wage leads to a statistically significant 2 percent decline in earnings for new firms, with the exception of the baseline earnings model, where new firms show a weakly significant 1 percent increase (Tables V-III, Column 7). Existing firms of all types, whether higher-earning, larger, older, or part of a chain, show increases in earnings relative to new firms, effectively offsetting the decline. This pattern suggests that in high-income areas, new firms are uniquely vulnerable to downward pressure on earnings in the wake of minimum wage increases. One possible explanation is that new firms may be attempting to undercut more established competitors by hiring workers with fewer credentials or less experience to keep costs low.

Entry Probability

I find evidence that chains and larger firms are more likely to enter the market following minimum wage increases in high-income counties, with a 10 percent increase in the minimum

wage associated with 0.3 and 0.2 percentage point increases in their entry rates, respectively (Table IX). Given approximately 77,000 childcare establishments and baseline entry rates of 12 to 13 percent, these effects imply that a 10 percent minimum wage hike could lead to roughly 385 additional chain or large firm entries nationwide in high-income counties. There is no significant effect in low-income counties, which may suggest that these firms anticipate greater success or profitability in higher-income areas.

Together, these findings point to meaningful heterogeneity in how establishments respond to minimum wage increases. Consistent with prior research, establishments in low-income counties tend to adjust on the extensive margin (e.g., exits), while those in high-income counties respond more on the intensive margin (e.g., employment and earnings). Across both contexts, the effects are most pronounced for new firms and existing firms that are younger, smaller, or lower-earning, suggesting these establishments may be more constrained by rising labor costs. In contrast, older, larger, and higher-earning firms appear more resilient, either because they are better able to absorb wage increases or because they are less affected by them. In high-income counties, minimum wage increases are also associated with increased entry by chains and larger firms, suggesting that these providers may view higher-income markets as more favorable environments for expansion under rising wage floors. Taken together, the exit and entry behavior across establishment types could suggest a broader change in the composition of the early childcare market.

Minimum Wage Impacts on Employment, Wages, and Firm Dynamics in the Self-Employed Sector

Unlike the waged sector, economic theory offers less consensus on how minimum wage policies affect the self-employed. In a dual-sector competitive model, where one sector is covered by the minimum wage and the other is not, two competing hypotheses emerge. One predicts that if minimum wage increases reduce employment in the waged sector, displaced workers may shift into self-employment, increasing labor supply and competition, thereby lowering earnings. An alternative hypothesis suggests that higher wages in the waged sector could attract self-employed workers to seek wage-paying jobs, decreasing labor supply in the unbound sector and potentially raising earnings for those who remain.

The self-employed childcare sector differs substantially from its waged counterpart. The average county has 55 self-employed providers, with owners earning just $9,965 annually (Table X). Turnover is much higher: entry and exit rates average 40 percent, compared to just 10 percent in the waged sector. Demographically, self-employed providers are less likely to be white and more likely to have a high school education or less (Figure II). Importantly, self-employed establishments are not directly covered by minimum wage laws since they report no employees, allowing owners to set their own compensation. Nevertheless, because self-employed providers operate in the same market as waged providers, they may be indirectly affected by minimum wage policy through changes in competition, demand, or worker composition.

To assess these spillover effects, I examine the relationship between minimum wage increases and outcomes in the self-employed sector. Both county- and establishment-level models show that a 10 percent increase in the minimum wage is associated with a statistically significant 2.5 to 3 percent decline in average income, and a 0.4 to 0.6 percentage point increase

in the exit rate (Table X and XI). While this exit effect represents only a 1 percent increase from the baseline exit rate of 40 percent, it suggests that higher minimum wages may slightly increase pressure on self-employed providers. There is no evidence of a corresponding change in the number of establishments or in the entry rate.

These patterns differ somewhat across county income levels. In high-income counties, the effects on earnings and exits mirror those in the overall self-employed sector. Earnings fall by roughly 2.8%, or a reduction of about $284 on average. Exits, on the other hand, increase by about 0.6 percentage points, which translates to an increase of about 1.5% from baseline. In low-income counties, exits also rise (by approximately 0.5 percentage points), but the estimate is less precise and not statistically significant (Table XII).[27]

One potential mechanism behind these changes is a shift in the composition of workers. As wages rise in the waged sector, more educated workers in the self-employed sector may transition into wage-paying jobs, while displaced or lower-skilled workers from the waged sector may enter self-employment. This inflow of less-educated workers could depress average earnings if they charge lower rates or face weaker demand. Consistent with this hypothesis, I find that a 10 percent increase in the minimum wage is associated with a small increase in the share of self-employed providers with a high school degree or less, about 1 percentage point in low-income counties and 0.5 percentage points in high-income counties (Table XIII). High-income counties also show a modest 0.4 percentage point decline in the share of providers with a bachelor's degree or more. These changes are marginally significant (at the 10 percent level) and

[27] County-level regression results by county income-status for the self-employed sector are similar and are available in Appendix G.

should be interpreted cautiously, especially given that worker education is measured at a single point in time and may be subject to measurement error.[28]

Taken together, these results suggest that the self-employed sector, while formally outside the reach of minimum wage policy, may still be indirectly affected. Increases in the minimum wage appear to reduce income and raise exit rates, particularly in high-income counties, possibly reflecting changes in market competition or worker composition. These findings underscore the interconnectedness of the waged and self-employed sectors and suggest that policy changes in one sector may have ripple effects in the other.

5. Conclusion

This study provides the first causal evidence on how minimum wage increases affect the structure of the U.S. childcare industry, a dual-sector labor market comprising both waged and self-employed providers. Using detailed county- and establishment-level data, I estimate the effects of minimum wage changes on employment, earnings, entry, and exit, highlighting how responses vary across establishment characteristics and regional income levels. While I find no significant effect on the overall number of childcare establishments or workers at the national level, the results point to meaningful heterogeneity in how individual establishments respond. I find establishments in low-income counties are more likely to adjust on the extensive margin, through increased exit, while those in high-income counties tend to adjust on the intensive

[28] These models are run on the sample of self-employed owners where education is reported and not-imputed and age is greater than 24. To show this sample is not meaningfully different than the overall sample, I rerun the overall results with this smaller sample. The results are similar, though slightly attenuated (see Appendix H).

margin, reducing employment. These patterns are most pronounced among new, smaller, lower-earning, and non-chain firms, while older, larger, and higher-earning firms appear more resilient.

In low-income counties, I also find that higher minimum wages are associated with increased entry of larger centers and chains, suggesting a potential shift toward consolidation. While these firms may offer more stability, such changes could alter local availability of care. In the self-employed sector, which is not directly bound by the minimum wage, I find modest spillover effects, particularly a compositional shift toward less credentialed providers, indicating that labor market dynamics in the waged sector may influence adjacent segments of the industry.

These findings are especially relevant as minimum wage policies continue to be adopted and expanded across U.S. states and cities, with many jurisdictions implementing wage schedules that raise the minimum wage over multiple years. In the early childhood care sector, where labor costs account for approximately 70 percent of total expenses, and where providers face regulatory limits on adjusting inputs such as child-teacher ratios, wage increases can introduce significant financial pressure. Roughly one-third of childcare workers earn within 10 percent of the minimum wage, making them particularly susceptible to wage policy shifts. As a result, increases in the wage floor may lead to unintended consequences. In low-income areas, where providers are more likely to exit, even small increases in closures could reduce access to care, particularly if new entrants do not fully replace those that exit or if incoming providers are concentrated in different neighborhoods or serve different populations. In high-income counties, by contrast, where establishments are more likely to remain open but reduce staffing, this could increase workloads for remaining staff or tighten child-teacher ratios if providers had previously operated above minimum standards. These adjustments may have ramifications for both availability and quality of care. Together, the findings underscore the importance of

understanding both the direct and indirect effects of labor market policies on a sector that plays a central role in child development and parental labor force participation.

Importantly, this study does not attempt to evaluate whether the benefits of minimum wage policies outweigh the costs. Rather, it identifies which types of establishments are most affected and in which settings effects are most pronounced. By doing so, it contributes new evidence to inform ongoing efforts to improve compensation in the early childhood sector without compromising access or quality. As policymakers weigh strategies to support both childcare workers and working families, understanding the distributional consequences of labor market policies will be critical to designing effective solutions.

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Appendix

Appendix A. Wage and Employment Estimates from Contemporary Studies

The below table and figure contain estimates for wage and employment outcomes from contemporaneous minimum wage studies across various time periods, low-wage industries, and datasets.

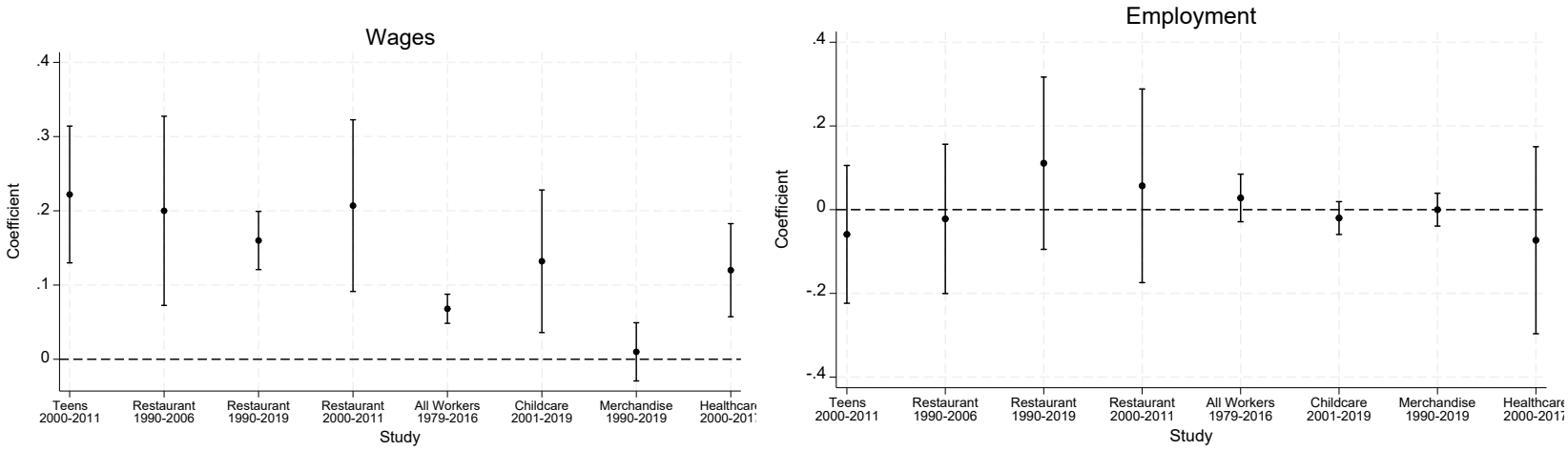
Table A.1 Wage and Employment Elasticity Estimates from Contemporary Studies

Outcome	Coefficient	SE	Dataset	Years	Group	Cite
Wage	0.222	0.047	Quarterly Workforce Indicators	2000-2011	Teens	Dube et al. 2016
Wage	0.207	0.059	Quarterly Workforce Indicators	2000-2011	Restaurant workers	Dube et al. 2016
Wage	0.132	0.049	Quarterly Workforce Indicators	2001-2019	Childcare industry	Brown and Herbst, 2023
Wage	0.200	0.065	Quarter Census of Employment and Wages	1990-2006	Restaurant workers	Dube et al. 2010
Wage	0.068	0.010	NBER Merged Outgoing Rotation Group of the Current Population Survey	1979-2016	All workers	Cengiz et al. 2019
Wage	0.160	0.020	Quarterly Workforce Indicators	1990-2019	Restaurant workers	Wursten and Reich, 2023
Wage	0.010	0.020	Quarterly Workforce Indicators	1990-2019	General merchandise and grocery stores	Wursten and Reich, 2023
Wage	0.120	0.032	Quarterly Workforce Indicators	2000-2017	Healthcare, workers with HS degree or less	Ruffini 2022
Employment	-0.059	0.084	Quarterly Workforce Indicators	2000-2011	Teens	Dube et al. 2016
Employment	-0.022	0.091	Quarterly Workforce Indicators	2000-2011	Restaurant workers	Dube et al. 2016
Employment	0.111	0.105	Quarterly Workforce Indicators	2001-2019	Childcare industry	Brown and Herbst, 2023
Employment	0.057	0.118	Quarter Census of Employment and Wages	1990-2006	Restaurant workers	Dube et al. 2010
Employment	0.028	0.029	NBER Merged Outgoing Rotation Group of the Current Population Survey	1979-2016	All workers	Cengiz et al. 2019
Employment	-0.020	0.020	Quarterly Workforce Indicators	1990-2019	Restaurant workers	Wursten and Reich, 2023
Employment	-0.000	0.020	Quarterly Workforce Indicators	1990-2019	General merchandise and grocery stores	Wursten and Reich, 2023
Employment	-0.073	0.114	Quarterly Workforce Indicators	2000-2017	Healthcare, workers with HS degree or less	Ruffini 2022

Note: Minimum wage elasticities from compiled contemporaneous studies.

Appendix A (continued). Wage and Employment Estimates from Contemporary Studies

Figure A.1 Wage and Employment Elasticity Estimates from Contemporary Studies



Appendix B: Event Study and Stacked Event Study Model Sensitivity Test

Estimating event study models allows me to test the key parallel trends assumption of the county-border discontinuity design while examining whether the minimum wage has a differential impact over time. While I have provided an aggregate view of the minimum wage's effect on labor market outcomes, short-term and long-term effects may differ as the market adjusts. For example, an initial wage hike could reduce labor availability, but over time, higher wages might stabilize the workforce or attract new providers. To capture these dynamic effects, I use a continuous treatment distributed-lag model (Schmidheiny and Siegloch 2020), which extends the logic of an event study. This approach multiplies event dummies by the size of the minimum wage change, allowing me to assess how a percentage change in the minimum wage influences outcomes over time. This results in the following model:

$$y_{ecpst} = \sum_{r=-25}^{25} \gamma_r \, I_{st}(\Delta lnMW_{s,t-r}) + \gamma Z_{ct} + \theta_c + \delta_{pt} + \partial_e + \varepsilon_{ecpst} \qquad (3)$$

In equation (3), I multiply each minimum wage event indicator, I_{st}, by the change in the logged minimum wage, $\Delta lnMW_{s,t-r}$. This model is fully saturated, examining the full spectrum of changes over the 25-year period. The full set of leads and lags is collinear with itself, therefore I set the reference period to the event period right before the focal minimum wage change, when r is -1. The coefficients of interest are γ_r, which describe the effect of the minimum wage for each period t-r before and after the minimum wage. This model continues to incorporate county, θ_c, pair time, δ_{pt}, and establishment, ∂_e, fixed effects, an error term, ε_{cpst}, as well as two-way clustered standard errors. Each γ_r will provide the estimate of a 1% increase in the minimum wage on the outcome of interest r years before or after the policy went into effect. As currently specified, the model indicates I will look at 25 years before and after a wage shift. I explore whether other specifications better fit the data.

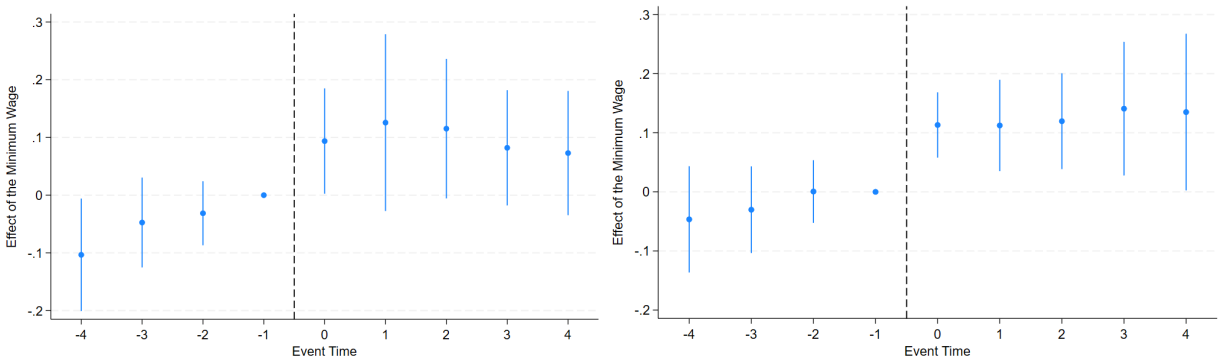
Up to this point, I have relied on methods that could be biased due to variations in the timing of minimum wage laws and the gradual nature of minimum wage adjustments. Minimum wage policies have the following features: staggered treatment adoption, non-absorbing treatment (meaning units can enter and exit treatment multiple times), and continuous treatment levels. This could lead to "bad comparisons" when the model uses earlier-treated units as a control group for later-treated units. This can lead to negative weighting in traditional two-way fixed effects models (TWFE) (Chaisemartin and D'Haultfoeuille, 2020). It can also result in biased difference-in-difference estimates if there is heterogeneity in cohorts (i.e., a group of units receiving treatment in the same period) (Sun and Abraham, 2021). To overcome these issues, researchers use stacked event study models (Jakiela 2021; Wursten and Reich, 2023; Cengiz et al. 2019). I run a version of a stacked event study to determine how sensitive the initial dynamic treatment effect is to heterogeneity concerns. Specifically, I estimate model (4).

$$y_{ecpsti} = \sum_{r=-4}^{4} \gamma_r \, I_{sti}(\Delta ln \, MW_{sti}) + \mu Z_{ct} + \varphi \omega_{cti} + \theta_{ci} + \delta_{pti} + \varepsilon_{ecpsti} \qquad (4)$$

This specification is very similar to (3) with a few key changes. First, in the stacked model, each event, i, counts as its own experiment. The resulting dataset will represent the full duplicated dataset described above, repeated i times for each event. I will separately estimate equation (4) for each event i, which is represented by a single state's minimum wage increase in time t, or MW_{sti}. All other states will serve as the control group. Since other states could experience minimum wage changes during the event window, I include ω_{cti}, which includes the cumulative minimum wage increase for small events and large events, separately, over the event window, excluding the focal event i for focal state s. Other studies consider stacked models that treat all minimum wage changes as an event regardless of their size as well as models that limit to only "large" minimum wage changes where the change is greater than a certain amount. I

include all changes, but I will consider running models with only prominent changes above a certain threshold to examine sensitivity. This model also includes county and county pair-time fixed effects, though notably these are now interacted by event. Similarly, I use two-way clustered standard errors for state by event and state border segment by event. I restrict each stack to only the 4 periods before and after the event.[29]

Figure B.1. Event study (left) and stacked event study (right) of logged average monthly earnings on the change in state minimum wages using Quarterly Workforce Indicators, 2001-2019



The figures show that the event study and stacked event study estimates yield highly similar results, both in magnitude and timing of effects. While the stacked event study coefficients are slightly more precise, the overall patterns are nearly identical. This consistency suggests that the identifying variation in minimum wage changes is not substantially affected by the sources of bias highlighted in recent critiques of two-way fixed effects difference-in-differences models (e.g., heterogeneity in treatment timing or treatment effects). In this context, the structure and timing of policy changes appear to align well with the assumptions underlying

[29] To ensure the timeframe is similar among treated and control groups, I choose the 8-year window surrounding the treated state's event time. If a treated state was treated in 2004, I will include all comparison units between 2000 and 2008. I made this decision partly for practical reasons. When creating a stacked dataset with over 300 events and a dataset of 140,000 observations, the resulting dataset grows tremendously slowing computation time. Moreover, by examining the period closer to the event time, I am restricting the period to a cleaner comparison group.

traditional difference-in-differences designs. As a result, the conventional estimates are likely to provide valid and unbiased estimates of the treatment effect.

Appendix C: Data Overview and Dataset Creation Description

Data Overview

The LBD, used by the United States Census Bureau's Center for Economic Studies to track establishments over time, allows researchers to explore business dynamics. It is created from the Census Bureau's Employer Business Register which is maintained by combining records from administrative data, federal business tax records from IRS and Social Security Administration, and Census Bureau collections, including surveys and semi-decadal Economic Censuses. It represents all private, non-farm business establishments with positive employment that operate in the United States, which represents roughly 80 percent of all employees. It includes unique longitudinal establishment and firm IDs, industry codes, establishment and firm birth and death dates, annual and quarterly payroll, employment as of March 12th, business names and address. Unlike the QCEW which is focused on accurately measuring real-time employment and wages, the Longitudinal Business Database's goal is to track businesses over time which is inherently tricky as establishments and firms can change their employer identification numbers over time as they merge, separate, or temporarily close (for detailed documentation see Chow et al., 2021; Jarmin & Miranda, 2002). The Integrated Longitudinal Business Database is an extension of the Longitudinal Business Database containing all non-employer, private businesses with data collected from payroll tax records, corporate and individual income tax returns, applications for an Employer Identification Number, and Census Bureau business surveys (Davis et al. 2006). It includes many of the same features of the LBD including establishment ID, industry code, income, and address, and can be used to track businesses as they move from non-employer to employer status. While the ILBD is an establishment-level dataset, many of the establishments are self-employed individuals, therefore these individuals' establishment IDs are equivalent to

their Protected Identification Key (PIK), or an individual ID. The ILBD has been consistently reporting annual employment statistics since 1977 with data presently available through 2022.

Dataset Creation Description

Data for this project was compiled from several sources. The matching process proceeds as follows. I began by taking the Longitudinal Business Database (LBD), which is a yearly census of businesses in the United States created from both surveys and administrative records. I appended each yearly LBD file from 1994 to 2021 keeping all records for NAICS code 624410 (Child Day Care Services). I then connected the County Business Patterns Business Register on the business' unique identifier to get the business' name and address. The LBD contains a unique identifier for the establishment (i.e., each individual provider location) and the firm (i.e., the parent company should a provider be part of a larger chain) as well as information on the establishments' annual and quarterly total payroll (in $1,000 of dollars) and total employment. This results in a yearly establishment-level file. Although the Census provides an estimate of the establishment opening and closing date, these do not always take into consideration businesses that close temporarily which is common in the childcare industry. Therefore, I impute temporary establishment closings and openings. For each firm, I create an indicator for whether the firm opened or closed in a particular year. Firms are only considered active if they have at least 1 employee and positive yearly payroll. A firm is considered to open in the first year they are seen in the data, unless that year is 1994 which is the first year of my data access. A firm is considered to close in year t if the firm is not active in year t+1 (i.e., if the firm had employment of at least 1 person in 2006 but is no longer seen in 2007). This decision was made because establishment-level employment is defined as of March 12 of the current year. This means, there is a high likelihood that the firm closed sometime during the year. I then do the reverse to determine an establishment's entry. An establishment is defined as entering the sample if they are not seen in year t-1 but in year t they have positive employment and payroll. Importantly, an establishment can enter and exit the sample in the sample year. I then merge in the Bureau of Labor Statistics CPI-U and transform all salaries to 2021

dollars.[30] I then merge in state and substate yearly minimum wage data. Before merging this data, I craft a list of every city that experiences a minimum wage shift in my sample and hand clean the raw city names in the LBD to improve the match. I determine all of the county-pairs for the difference-in-difference border discontinuity model by taking the Census' 2010 County Adjacency file, limiting the dataset to only counties on state borders, and then creating a county pair ID (e.g., New York county Chautauqua and Pennsylvania county Erie are bordering counties and would receive the same county pair ID to identify that these two counties are in fact neighbors). I merge an indicator for whether an establishment's county is a state border county and the unique county pair identifier. An example of the final file is below.

Table. Example table of the Provider-Level Dataset created from the LBD

ID	Year	County	Business Name	County Pair ID	Employees	Years Active	Closed	Opened	Minimum Wage
1	2014	001	ABC Care	1	5	1	.	1	7.25
1	2015	001	ABC Care	1	6	2	0	0	7.50
1	2016	001	ABC Care	1	5	3	1	.	7.50
2	1994	001	XYZ Care	2	12	1	0	0	7.25
2	1995	001	XYZ Care	2	10	2	1	.	7.25

NOTE: All data is made up. The purpose of this table is to illustrate how the analytic dataset is constructed for key outcomes of interest.

I repeat the process outlined above for the LBD, to create a panel of self-employed providers who fill out the Schedule C IRS tax form. The data for these individuals is held in the Integrated Longitudinal Business Database (ILBD). I keep all records for providers associated with NAICS code 624410 (Child Day Care Services). Like above, I merge in state minimum wage and link the county adjacency file.[31] I

[30] I use CPI for All Items for January of the relevant year.
[31] ILBD contains the business address but not their city. I am currently working on backing out the city with geocoding. Once this is done substate minimum wage data will be added.

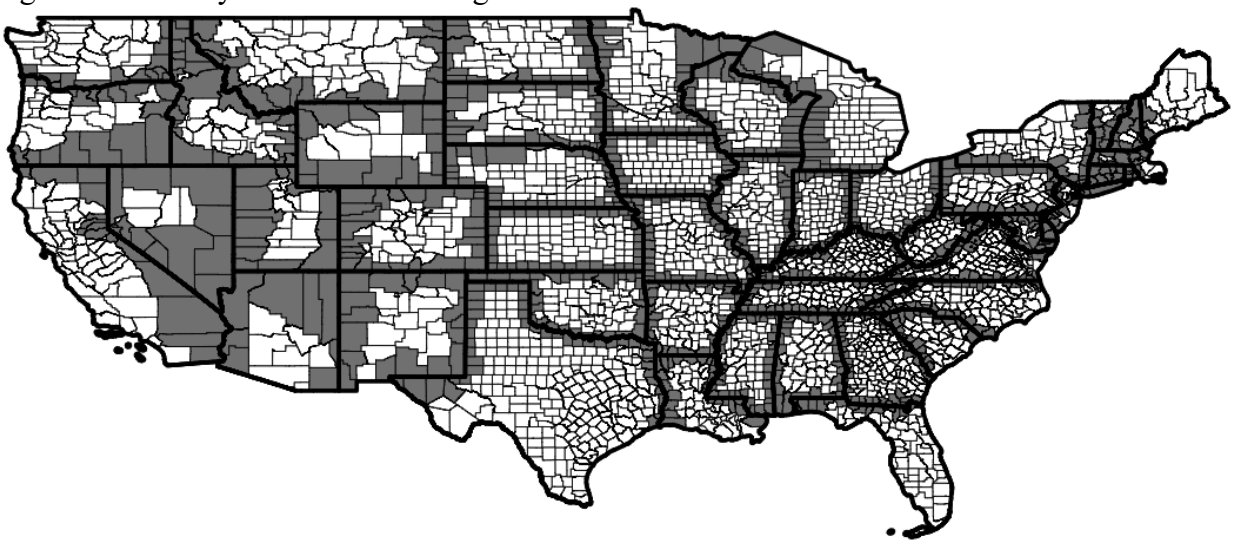
adjust all incomes to 2021 dollars using CPI-U. Unlike the LBD dataset, these providers are also individuals, so I merge in their demographic characteristics from the LEHD demographic file and their survey responses to the ACS. I am working to determine the percent of workers in the LEHD demographic file that have a record in the ILBD suggesting that most workers who are self-employed were at one time employed by a business participating in the state Unemployment Insurance system. I then create similar metrics to those above for ILBD openings and closings. The ILBD has more individuals coming in and out of the sample over time (i.e., a person may appear in 2003, they will be missing in 2004, and then they will come back in 2005). This time skipping pattern could be real, or it could be an artifact of people not always filing their taxes. I assume all individuals in the ILBD belong to the family day home provider sector, since they are self-employed. To validate this claim I do two things. First, I merge in the LEHD residential addresses on unique person identifier, PIK, and year as well as the Master Address Extract File (though notably this is only available starting in 2008). I then calculate the percent of workers who are living and working within 1 mile of their residence. This phase of the project is ongoing.

While the LBD, ILBD, and LEHD data are restricted access data, I do make available a version of the analysis using all publicly available data to ensure reproducibility. Specifically, I make available code to create the underlying dataset including: 1) covariates, 2) border counties including all relevant information such as centroid distance, county-pair and county-pair-time IDs, state border segments IDs, and 3) wage and employment estimates from the Quarterly Workforce Indicators dataset.

Appendix D: Map of the United States County Borders

The below map shows United States counties and highlights all border counties in dark gray.

Figure D.1 County Border Pairs Along State Borders



Source: Census Tiger Shapefile for States and Counties and Census County Adjacency Files

Appendix E. County-Level Sensitivity Models

The preferred model is a cross-state county border discontinuity with 1) controls for logged population, employment, and pre-kindergarten enrollment, 2) county and pair-time fixed effects, 3) two-way state and state border clustered standard errors, and 4) border-population weights. I run a series of sensitivity tests including: Panel A. removing pre-kindergarten enrollment due to the concern this might be endogenous (i.e., impacted by the minimum wage), Panel B. limiting to only county pairs whose centroids are within 75km to avoid the concern that larger counties may not be good comparison groups due to dissimilar labor markets, Panel C. adjusting from a border weighted population weight to a traditional population weight, and Panel D. adding recessionary controls, defined as state by recession year dummies (2007, 2008, and 2009), to control for the fact that the recession may have disrupted the early childcare labor market. The findings from the preferred model are very similar to each of these adjustments, suggesting the minimum wage is the driving cause of the effects.

Table E.1 County-Level Sensitivity Tests for the Waged Sector

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total Establishments (Log)	Total County Employment (Log)	Workers per Establishment (log)	First Quarter Payroll per Employment (Log)	Payroll per Establishment (Log)	Exit Rate (0/1)	Entry Rate (0/1)
Panel A: Preferred Model with No Pre-K Enrollment Control							
Ln(MW)	-0.021	-0.108	-0.355	0.039	-0.004	0.029**	0.025
	(0.107)	(0.129)	(0.249)	(0.054)	(0.133)	(0.012)	(0.021)
Panel B. Preferred Model Limited to Counties Within 75km							
Ln(MW)	-0.017	-0.123	-0.379	-0.005	-0.05	0.031**	0.011
	(0.119)	(0.146)	(0.267)	(0.063)	(0.144)	(0.012)	(0.021)
Panel C. Preferred Model with Traditional Population Weights							
Ln(MW)	0.018	-0.147	-0.456*	0.032	-0.085	0.032**	0.033
	(0.094)	(0.117)	(0.268)	(0.053)	(0.139)	(0.013)	(0.022)
Panel D. Preferred Model and Recessionary Controls							
Ln(MW)	-0.016	-0.144	-0.448*	0.017	-0.05	0.034***	0.037*
	(0.112)	(0.142)	(0.250)	(0.057)	(0.135)	(0.013)	(0.022)

NOTE: Based on author's calculations using restricted-access Census micro-data. Number of observations is 54,500 in panels A, C, and D and 45,000 in Panel B. All variables are naturally logged,

with the exception of exit and entry which are indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. They are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693) (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

Table E.2 County-Level Sensitivity Tests for the Self-Employed Sector

	(1)	(2)	(3)	(4)
	Total Establishments (ASINH)	Exit Rate (0/1)	Entry Rate (0/1)	Income (ASINH)
Panel A. Preferred Model with No Pre-K Enrollment Control				
Ln(MW)	0.123	0.038**	0.000	-0.320**
	(0.158)	(0.017)	(0.030)	(0.135)
Panel B. Preferred Model Limited to Counties Within 75km				
Ln(MW)	0.197	0.048***	0.002	-0.398***
	(0.173)	(0.017)	(0.029)	(0.137)
Panel C. Preferred Model with Traditional Population Weights				
Ln(MW)	0.156	0.041**	0.005	-0.283**
	(0.162)	(0.017)	(0.031)	(0.133)
Panel D. Preferred Model and Recessionary Controls				
Ln(MW)	0.155	0.032*	0.002	-0.288*
	(0.174)	(0.016)	(0.029)	(0.150)

NOTE: Based on author's calculations using restricted-access Census micro-data. Number of observations is 65,000 in panels A, C, and D for outcome 1 and 64,500 for outcomes 2-4. For Panel B the number of observations is 52,00 and 51,500 for outcomes 1 and 2-4, respectively. Number of observations is 45,000 in Panel B. Total establishments and income are transformed using inverse hyperbolic sine and exit and entry are indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. They are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693) (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

Appendix F. County-Level Results by Income for the Waged and Self-Employed Sectors

Table F.1 County-Level Regression Results by County Income Status for the Waged Sector

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total Establishments (Log)	Total County Employment (Log)	First Quarter Payroll per Employment (Log)	Workers per Establishment (Log)	Exit Rate (0/1)	Entry Rate (0/1)	Payroll per Establishment (Log)
Panel A: Low-Income Counties							
Ln(MW)	0.009	-0.412	0.419**	-0.349	0.096*	0.029	-0.157
	(0.218)	(0.325)	(0.160)	(0.308)	(0.051)	(0.076)	(0.424)
R-squared	0.974	0.962	0.764	0.810	0.577	0.586	0.868
Observations	20,500	20,500	20,500	20,500	20,500	20,500	20,500
Raw Mean	1.6	3.5	8.2	2.5	0.1	0.1	11.6
Panel B: High-Income Counties							
Ln(MW)	0.022	-0.072	0.021	-0.431*	0.025*	0.022	-0.055
	(0.108)	(0.128)	(0.059)	(0.254)	(0.013)	(0.027)	(0.131)
R-squared	0.996	0.994	0.909	0.886	0.653	0.639	0.941
Observations	20,500	20,500	20,500	20,500	20,500	20,500	20,500
Raw Mean	2.7	4.9	8.3	2.5	0.1	0.1	11.9

NOTE: Based on author's calculations using restricted-access Census micro-data. All variables are naturally logged, with the exception of exit and entry which are indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. They are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693) (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

Appendix G. County-Level Regression Results by County Income Level for Self-Employed

Table G.1 County-Level Regression Results by County Income Level for Self-Employed Sector

	(1)	(2)	(3)	(4)
	Total Establishments (Asinh)	Exit Rate (0/1)	Entry Rate (0/1)	Income (Asinh)
Panel A: Low-Income Counties				
Ln(MW)	0.187	0.049	0.029	0.059
	(0.319)	(0.059)	(0.070)	(0.142)
R^2	0.983	0.658	0.662	0.787
Raw Mean	4.2	0.4	0.4	9.8
Panel B: High-Income Counties				
Ln(MW)	0.224	0.046***	0.004	-0.381**
	(0.164)	(0.016)	(0.029)	(0.153)
R^2	0.994	0.905	0.911	0.932
Raw Mean	5.4	0.4	0.3	10.1

NOTE: Based on author's calculations using restricted-access Census micro-data. Sample size is 25,000 in Panel A and 23,000 in Panel B. Total establishments and income are both transformed using the inverse hyperbolic sine. Exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Models are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693). Two-way state and state border cluster robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

Appendix H. Self-Employed Establishment-Level Sensitivity Models

The following table is a sensitivity test, to see if the regression results hold when limiting to self-employed owners who non-imputed education and age over 24. The following results show that in general, the estimated effects are similar in sign but are slightly attenuated likely due to measurement bias in how education is calculated.

Table H.1 Establishment-Level Regression Results for the Self-Employed Childcare Industry, limited to non-imputed education and age over 24

	(1)	(2)	(3)	(4)	(5)	(6)
	Income (asinh)		Exits (0/1)		Entry (0/1)	
	Low-Income	High-Income	Low-Income	High-Income	Low-Income	High-Income
Ln(MW)	0.045	-0.179	0.00272	0.0261	0.0162	0.0144
	(0.164)	(0.134)	(0.0529)	(0.0244)	(0.0511)	(0.0221)
R-squared	0.086	0.065	0.054	0.036	0.072	0.051
Observations	607,000	2,413,000	607,000	2,413,000	607,000	2,413,000

NOTE: Based on author's calculations using restricted-access Census micro-data. Income is transformed using inverse hyperbolic sine and exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. They are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693) (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

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Table I. Characteristics of All US Counties and Border Counties

	All Counties			Border Counties			
	Mean	Standard Deviation	Number	Mean	Standard Deviation	Number	P-value
County Characteristics							
Population: Total	104.94	334.32	3142	102.58	282.91	1184	0.83
Population: 0-4 Year Olds	6.18	20.27	3142	6.00	17.21	1184	0.79
Population: 20-64 Year Olds	61.69	203.26	3142	60.26	171.04	1184	0.83
Median Household Income (per $1000)	58.94	15.27	3142	58.69	15.54	1184	0.62
Rural (%)	53.15	49.91	3140	54.14	49.85	1184	0.56
Childcare Firms							
# Establishments	25.83	80.25	2834	27.07	80.09	1054	0.67
Average Weekly Wage	231.24	211.99	2834	231.27	214.08	1054	1.00
# Employees	804.55	2521.20	2834	859.60	2682.68	1054	0.55

NOTE: All statistics based on author's calculations using 2021 data, except for rural which is a 2013 statistic. Population statistics are from the ACS 5-year statistics and are per 1,000 residents, Unemployment Rate is from the Bureau of Labor Statistics. Median Income is from U.S. Department of Commerce, Bureau of the Census, Small Area Income and Poverty Estimates (SAIPE) Program. All childcare firm characteristics are from the publicly available Quarterly Census of Employment and Wages. The p-value is associated with a difference in means test between the all and border county samples.

Table II. County-Level Regression Results for the Waged Sector

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Total Establishments (asinh)	Total Establishments (ln)	Total County Employment (asinh)	Total County Employment (ln)	Payroll per Establishment (asinh)	Payroll per Establishment (ln)	Workers per Establishment (ln)	Exit Rate (0/1)	Entry Rate (0/1)	First Quarter Payroll per Employment (ln)
Ln(MW)	-0.001	-0.021	-0.130	-0.108	0.155	-0.004	-0.396*	0.037***	0.031	0.039
	(0.102)	(0.107)	(0.122)	(0.129)	(0.166)	(0.133)	(0.229)	(0.011)	(0.022)	(0.063)
R-squared	0.995	0.995	0.990	0.992	0.832	0.930	0.876	0.617	0.621	0.889
Observations	65,000	54,500	65,000	54,500	65,000	54,500	54,500	54,500	54,500	54,500
Raw Mean	2.5	2.1	4.4	4.2	11.6	11.7	2.5	0.1	0.1	8.2
Real Mean	6.1	8.2	40.7	66.7	$54,548.90	$120,571.71	12.2	10%	10%	$3,640.95

NOTE: Based on author's calculations using restricted-access Census micro-data. Employment is defined as of March 12 of the current year. First quarter income is defined as the first quarter payroll divided by enrollment. Ln(MW) is defined as the state's mean of the minimum wage during

the first quarter for all variables except payroll.[32] This is because all variables are based on values from the first quarter, with the exception of payroll which is yearly. Ln(MW) is constructed as the state average of the minimum wage for payroll. Once the individual establishment-level data is cleaned, the data is aggregated to the county-level. For counties with no employment, establishments, or payroll I impute those counties values as 0. To keep these counties in the estimation, I transform each using the inverse hyperbolic sine. For sensitivity, I also run these select outcomes dropping counties with no establishments and logging the outcomes instead. Exit (or entry) rates are continuous measures defined as the number of exits (or entry) in year t divided by the number of firms in year t-1. Counties with no lagged establishments are dropped from the regression. First-quarter income and establishment-level employment are both transformed using the natural log, meaning any county with no workers will not enter the model. First quarter income and payroll are all in real 2021 dollars. First quarter income, payroll, and firm-level employment are constructed as the median across all establishments in the county. First quarter income and payroll are weighted by establishment employment (meaning larger establishments will contribute more). All models are aggregated from establishment level data which is limited to the sample of establishments with positive enrollment and payroll. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. They are weighted using border population weights. Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693) (CBDRB-FY25-P2912-R12460).

[32] The independent variable depends on the construction of the outcome. Payroll, which is yearly, is measured using the yearly mean of minimum wage. All other statistics are calculated as of the first quarter of the current year, which is why they rely on a measure of the minimum wage as of the first quarter of the year.

Table III. Establishment-Level Regression Results for the Waged Establishments

	(1)	(2)	(3)	(4)	(5)
	Employment (ln)	First Quarter Payroll per Employment (ln)	Exits (0/1)	Entry (0/1)	Payroll per Establishment (ln)
Ln(MW)	-0.276**	-0.0132	0.0289**	0.00846	-0.308*
	(0.134)	(0.0432)	(0.0118)	(0.0181)	(0.168)
R^2	0.095	0.155	0.041	0.043	0.123
Raw Mean	1.98	8.33	0.10	0.11	11.7
Real Mean	7.24	$4,146.42	10%	11%	$120,571.71

NOTE: Based on author's calculations using restricted-access Census micro-data. Number of observations is 1,501,000. Employment, payroll, and first quarter income are naturally logged. Exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table IV. Establishment-Level Regression Results by Income Status for the Waged Establishments

	(1)	(2)	(3)	(4)	(5)
	Employment (ln)	First Quarter Payroll per Employment (ln)	Exits (0/1)	Entry (0/1)	Payroll per Establishment (ln)
Panel A: Low-Income Counties					
Ln(MW)	-0.185	0.386***	0.107**	0.0469	0.0676
	(0.200)	(0.115)	(0.0463)	(0.0443)	(0.321)
R-squared	0.164	0.181	0.089	0.096	0.183
Raw Mean	1.83	8.18	0.11	0.12	11.4
Real Mean	6.2	$3,568.85	11%	12%	$133,252.35
Panel B: High-Income Counties					
Ln(MW)	-0.307**	-0.0573	0.0243**	-0.00183	-0.393**
	(0.138)	(0.0456)	(0.0120)	(0.0206)	(0.169)
R-squared	0.078	0.136	0.032	0.033	0.102
Raw Mean	2.01	8.36	0.09	0.11	11.8
Real Mean	7.5	$4,272.69	9%	11%	$133,252.35

NOTE: Based on author's calculations using restricted-access Census micro-data. Panel A is based on a sample of 271,000 and Panel B is based on 1,229,000. Employment, payroll, and first quarter income are naturally logged. Exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table V. Establishment-Level Regression Results by Establishment Earnings for Waged Establishments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low-Income Counties				High-Income Counties			
	Exit (0/1)	Employment (ln)	First Quarter Payroll per Employment (ln)	Payroll per Establishment (ln)	Exit (0/1)	Employment (ln)	First Quarter Payroll per Employment (ln)	Payroll per Establishment (ln)
Ln(Minimum Wage)	0.117**	-0.285	0.325***	-0.127	-0.00478	-0.257**	0.1000*	-0.156
	(0.0463)	(0.200)	(0.103)	(0.287)	(0.0169)	(0.122)	(0.0523)	(0.142)
Existing, Lagged Earnings Below 10% Minimum Wage	-0.0899**	0.351**	-0.309***	-0.126	-0.154***	0.529***	-0.371***	0.138
	(0.0335)	(0.169)	(0.0733)	(0.206)	(0.0220)	(0.112)	(0.0627)	(0.118)
Existing, Lagged Earnings Above 10% Minimum Wage	-0.0327	-0.441	0.0618	-0.533	-0.157***	0.220*	-0.00549	0.195
	(0.0340)	(0.318)	(0.0928)	(0.391)	(0.0269)	(0.112)	(0.0765)	(0.171)

Ln(Minimum Wage) x Lagged Earnings Below 10% Minimum Wage	-0.022	0.0933	0.202***	0.371***	0.0166	0.0439	0.202***	0.223***
	(0.0191)	(0.0952)	(0.0428)	(0.119)	(0.0102)	(0.0620)	(0.0286)	(0.0633)
Ln(Minimum Wage) x Lagged Earnings Above 10% Minimum Wage	-0.0737***	0.705***	0.324***	1.083***	0.00183	0.397***	0.318***	0.686***
	(0.0191)	(0.190)	(0.0540)	(0.238)	(0.0128)	(0.0642)	(0.0403)	(0.0961)
R-squared	0.111	0.216	0.381	0.291	0.055	0.143	0.36	0.229
Raw Baseline Mean	0.23	1.28	7.96	10.60	0.21	1.30	8.11	10.90
Real Baseline Mean	23%	3.60	$2,854.07	$40,134.84	21%	3.67	$3,327.58	$54,176.36

Note: Based on author's calculations using restricted-access Census micro-data. Employment, payroll, and first quarter income are naturally logged. Exit is an indicator. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Low-income sample contains 271,000 individual establishments located in low-income counties and the high-income sample contains 1,229,000 located in high-income counties. These figures are rounded based on Census guidelines. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table VI. Establishment-Level Regression Results by Establishment Size for Waged Establishments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low-Income Counties				High-Income Counties			
	Exit (0/1)	Employment (Ln)	First Quarter Payroll per Employment (Ln)	Payroll per Establishment (Ln)	Exit (0/1)	Employment (Ln)	First Quarter Payroll per Employment (Ln)	Payroll per Establishment (Ln)
Ln(Minimum Wage)	0.128***	-0.251	0.174	-0.283	0.0101	-0.364***	-0.235***	-0.611***
	(0.0461)	(0.165)	(0.126)	(0.271)	(0.0163)	(0.0952)	(0.0634)	(0.128)
Existing, Lagged Employment (0-19)	-0.0775**	0.165	-0.211***	-0.209	-0.151***	0.328***	-0.104*	0.189
	(0.0328)	(0.126)	(0.0774)	(0.182)	(0.0213)	(0.0804)	(0.0529)	(0.129)
Existing, Lagged Employment (20+)	-0.0809*	1.351***	-0.260**	0.951*	-0.212***	1.473***	-0.133	1.375***
	(0.0435)	(0.392)	(0.126)	(0.530)	(0.0307)	(0.121)	(0.0835)	(0.177)
Ln(Minimum Wage) x Existing, Lagged Employment (0-19)	-0.0308	0.123	0.237***	0.426***	0.0143	0.0451	0.176***	0.199***

	(0.0188)	(0.0736)	(0.0479)	(0.110)	(0.0102)	(0.0431)	(0.0294)	(0.0699)
Ln(Minimum Wage) x Existing, Lagged Employment (20+)	-0.0608**	0.359	0.386***	0.816**	0.017	0.299***	0.291***	0.555***
	(0.0240)	(0.223)	(0.0805)	(0.307)	(0.0140)	(0.0749)	(0.0469)	(0.0996)
R-squared	0.112	0.421	0.217	0.401	0.058	0.423	0.173	0.389
Raw Baseline Mean	0.23	1.28	7.96	10.60	0.21	1.30	8.11	10.90
Real Baseline Mean	23%	3.60	$2,854.07	$40,134.84	21%	3.67	$3,327.58	$54,176.36

Note: Based on author's calculations using restricted-access Census micro-data. Employment, payroll, and first quarter income are naturally logged. Exit is an indicator. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Low-income sample contains 271,000 individual establishments located in low-income counties and the high-income sample contains 1,229,000 located in high-income counties. These figures are rounded based on Census guidelines. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table VII. Establishment-Level Regression Results by Establishment Age for Waged Establishments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low-Income Counties				High-Income Counties			
	Exit (0/1)	Employment (Ln)	First Quarter Payroll per Employment (Ln)	Payroll per Establishment (Ln)	Exit (0/1)	Employment (Ln)	First Quarter Payroll per Employment (Ln)	Payroll per Establishment (Ln)
Ln(Minimum Wage)	0.122**	-0.367	0.168	-0.404	0.00778	-0.422***	-0.234***	-0.667***
	(0.0466)	(0.221)	(0.133)	(0.352)	(0.0173)	(0.130)	(0.0640)	(0.159)
Existing, Lagged Age 1-3	-0.0256	0.201	-0.132*	-0.0738	-0.0929***	0.328***	-0.117***	0.183*
	(0.0346)	(0.165)	(0.0739)	(0.204)	(0.0284)	(0.0642)	(0.0404)	(0.0945)
Existing, Lagged Age 4 or Older	-0.0978***	0.12	-0.263***	-0.323	-0.187***	0.553***	-0.106	0.420**
	(0.0336)	(0.182)	(0.0906)	(0.251)	(0.0233)	(0.108)	(0.0650)	(0.161)
Ln(Minimum Wage) x Existing, Lagged Age 1-3	-0.0380*	0.0806	0.152***	0.289**	0.00504	0.031	0.143***	0.153**
	(0.0197)	(0.0977)	(0.0451)	(0.122)	(0.0151)	(0.0396)	(0.0239)	(0.0576)

Ln(Minimum Wage) x Existing, Lagged Age 4 or Older	-0.0325*	0.326***	0.302***	0.706***	0.0192*	0.169***	0.216***	0.363***
	(0.0188)	(0.111)	(0.0597)	(0.163)	(0.0106)	(0.0531)	(0.0356)	(0.0798)
R-squared	0.114	0.215	0.211	0.24	0.06	0.148	0.167	0.173
Raw Baseline Mean	0.23	1.28	7.96	10.60	0.21	1.30	8.11	10.90
Real Baseline Mean	23%	3.60	$2,854.07	$40,134.84	21%	3.67	$3,327.58	$54,176.36

Note: Based on author's calculations using restricted-access Census micro-data. Employment, payroll, and first quarter income are naturally logged. Exit is an indicator. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Low-income sample contains 271,000 individual establishments located in low-income counties and the high-income sample contains 1,229,000 located in high-income counties. These figures are rounded based on Census guidelines. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table VIII. Establishment-Level Regression Results by Establishment Chain-Status for Waged Establishments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low-Income Counties				High-Income Counties			
	Exit (0/1)	Employment (Ln)	First Quarter Payroll per Employment (Ln)	Payroll per Establishment (Ln)	Exit (0/1)	Employment (Ln)	First Quarter Payroll per Employment (Ln)	Payroll per Establishment (Ln)
Ln(Minimum Wage)	0.132***	-0.403**	0.16	-0.447	0.00985	-0.407***	-0.222***	-0.632***
	(0.0447)	(0.194)	(0.116)	(0.299)	(0.0173)	(0.137)	(0.0613)	(0.171)
Existing, Lagged Non-Chain	-0.0758**	0.166	-0.178**	-0.208	-0.151***	0.372***	-0.167**	0.164
	(0.0346)	(0.147)	(0.0688)	(0.191)	(0.0197)	(0.118)	(0.0639)	(0.176)
Existing, Lagged Chain	-0.104**	0.418	-0.0291	0.353	-0.195***	0.846***	0.0699	0.941***
	(0.0443)	(0.545)	(0.157)	(0.652)	(0.0377)	(0.173)	(0.0837)	(0.224)
Ln(Minimum Wage) x Existing, Lagged Non-Chain	-0.0311	0.196**	0.193***	0.474***	0.0122	0.149**	0.205***	0.335***
	(0.0194)	(0.0828)	(0.0468)	(0.117)	(0.00968)	(0.0617)	(0.0355)	(0.0926)

Ln(Minimum Wage) x Existing, Lagged Chain	-0.0431*	0.353	0.340***	0.698*	0.0174	0.166	0.195***	0.327**
	(0.0241)	(0.313)	(0.0908)	(0.384)	(0.0184)	(0.107)	(0.0472)	(0.130)
R-squared	0.112	0.229	0.267	0.28	0.055	0.152	0.178	0.187
Raw Baseline Mean	0.23	1.28	7.96	10.60	0.21	1.30	8.11	10.90
Real Baseline Mean	23%	3.60	$2,854.07	$40,134.84	21%	3.67	$3,327.58	$54,176.36

Note: Based on author's calculations using restricted-access Census micro-data. Employment, payroll, and first quarter income are naturally logged. Exit is an indicator. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Low-income sample contains 271,000 individual establishments located in low-income counties and the high-income sample contains 1,229,000 located in high-income counties. These figures are rounded based on Census guidelines. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table IX. Establishment-Level Regression Results by Establishment Characteristics for Waged Establishments

Panel A. Establishment Chain-Status

	(1)	(2)
	Entry (0/1)	
	County Income	
	Low	High
Ln(MW)	0.0487	-0.0145
	(0.0445)	(0.0200)
Chain	-0.0878	-0.122***
	(0.0524)	(0.0252)
Ln(MW) x Chain	0.00918	0.0342***
	(0.0300)	(0.0120)
R-squared	0.102	0.038
Observations	271,000	1,229,000
Raw Mean	0.13	0.12

Panel B. Establishment Size

	(1)	(2)
	Entry (0/1)	
	County Income	
	Low	High
Ln(MW)	0.0408	-0.00931
	(0.0461)	(0.0216)
Employment (20+)	-0.117*	-0.122***
	(0.0687)	(0.0130)
Ln(MW) x Employment (20+)	0.0148	0.0214***
	(0.0405)	(0.00767)
R- squared	0.104	0.044
Observations	271,000	1,229,000
Raw Mean	0.13	0.13

NOTE: Based on author's calculations using restricted-access Census micro-data. Reference group is new establishments in year t. All models include logged population, private employment, pre-K enrollment controls and county and county-pair time fixed effects. FSRDC Project Number

2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw baseline means which are reported as proportions. *** p<0.01, ** p<0.05, * p<0.1.

Table X. County-Level Regression Results for Self-Employed Childcare Industry

	(1)	(2)	(3)	(4)
	Total Establishments (Asinh)	Exit Rate (0/1)	Entry Rate (0/1)	Income (Asinh)
Ln(MW)	0.207	0.043***	0.011	-0.310**
	-0.158	-0.016	-0.029	-0.136
R-squared	0.995	0.825	0.835	0.908
Observations	65,000	64,500	64,500	64,500
Raw Mean	4.70	0.40	0.40	9.90
Real Mean	54.97	40%	40%	$9,965.19

NOTE: Based on author's calculations using restricted-access Census micro-data. Total establishments and income are both transformed using the inverse hyperbolic sine. Exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Models are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table XI. Establishment-Level Regression Results by County Income Level for Self-Employed Childcare Industry

	(1)	(2)	(3)
	Income (Asinh)	Exit (0/1)	Entry (0/1)
Ln(MW)	-0.249*	0.0554**	0.0316*
	(0.128)	(0.0247)	(0.0188)
R-squared	0.054	0.041	0.027
Raw Mean	9.92	0.41	0.41
Real Mean	$10,166.50	41%	41%

NOTE: Based on author's calculations using restricted-access Census micro-data. Total establishments and income are both transformed using the inverse hyperbolic sine. Exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. Models are weighted using border population weights. FSRDC Project Number 2912 (CBDRB-FY24-P2912-R11693). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table XII. Establishment-Level Regression Results for the Self-Employed Childcare Industry

	(1)	(2)	(3)	(4)	(5)	(6)
	Income (Asinh)		Exit (0/1)		Entry (0/1)	
	Low-Income	High-Income	Low-Income	High-Income	Low-Income	High-Income
Ln(MW)	-0.0904	-0.281*	0.0396	0.0625**	0.0553	0.0286
	(0.131)	(0.146)	(0.0433)	(0.0285)	(0.0527)	(0.0205)
R^2	0.065	0.048	0.027	0.027	0.042	0.04
Raw Mean	9.92	9.92	0.41	0.41	0.41	0.41
Real Mean	$10,166.50	$10,166.50	41%	41%	41%	41%

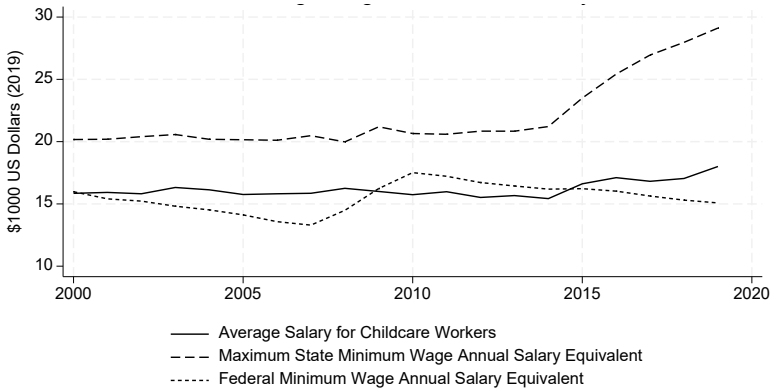
NOTE: Based on author's calculations using restricted-access Census micro-data. Observations for low-income counties is 2,696,000 and high-income counties is 11,240,000. The sample excludes 0.5% of establishments with extreme payroll defined as non-positive or greater than $200,000. Income is transformed using the inverse hyperbolic sine. Exit and entry are both indicators. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw (transformed) values and the real (back-transformed) values. *** p<0.01, ** p<0.05, * p<0.1.

Table XIII. Establishment-Level Regression Results by Owner Education for Self-Employed Owners

	(1)	(2)
	Low-income counties	High-Income Counties
High School or Less (0/1)	0.101*	0.0510*
	(0.0546)	(0.0263)
R-squared	0.072	0.049
Mean	0.56	0.46
Some College (0/1)	-0.0857	-0.0116
	(0.0649)	(0.0248)
R-squared	0.057	0.026
Mean	0.33	0.34
Bachelor's Degree or More (0/1)	-0.0154	-0.0394*
	(0.0486)	(0.0218)
R-squared	0.055	0.046
Mean	0.12	0.20

NOTE: Based on author's calculations using restricted-access Census micro-data. All models include logged population, private employment, pre-K enrollment controls as well as county and county-pair time fixed effects. FSRDC Project Number 2912 (CBDRB-FY25-P2912-R12460). Number of observations in low-income counties is N=607,000 and N=2,413,000 in high-income counties. Two-way state and state border cluster robust standard errors in parentheses. All means are reported as the raw values and reflect the proportion of owners with the respective education credential. *** p<0.01, ** p<0.05, * p<0.1.

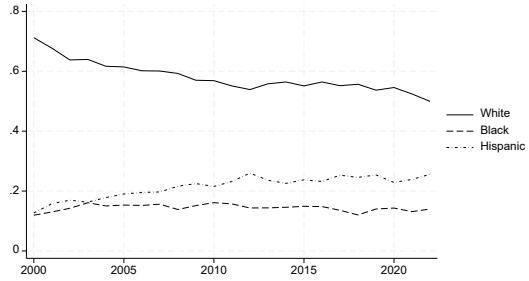
Figure I. Average Annual Salary for Childcare Workers Compared to Federal and State Minimum Wage Annual Salary Equivalents



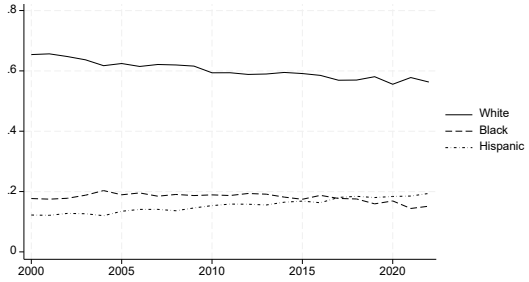
NOTE: Based on author's calculations using American Community Survey 1-year estimates.

Figure II. Composition of the Self-Employed and Waged Childcare Sectors

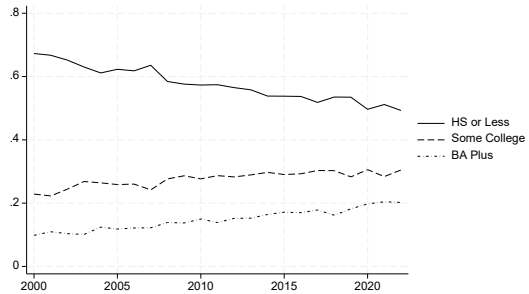
Panel A. Childcare Worker Race for Self-Employed Sector



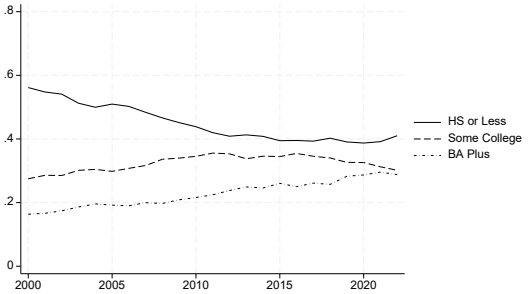
Panel B. Childcare Worker Race for Waged Sector



Panel C. Childcare Worker Education for Self-Employed Sector

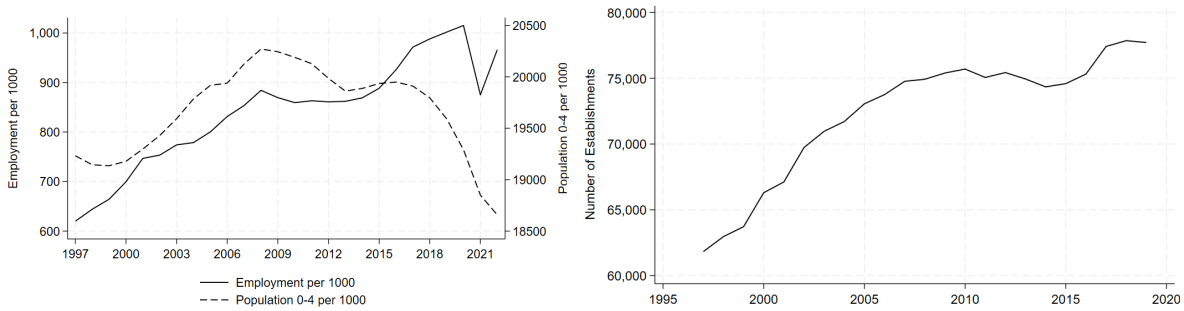


Panel D. Childcare Worker Education for Waged Sector



NOTE: Based on author's calculations using the American Community Survey 1-year estimates

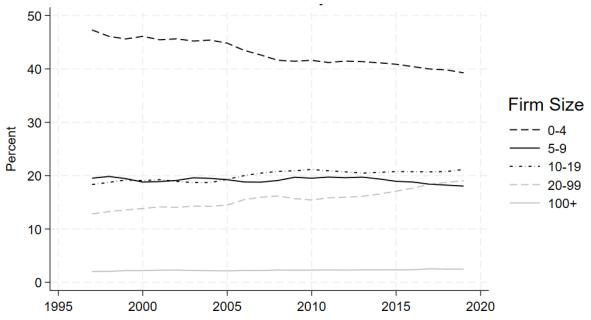
Figure III. Childcare Industry Employment and Population Trends (Left) and Total Childcare Establishments (Right) Over Time



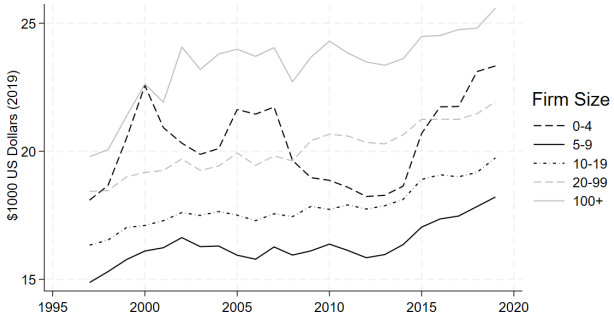
NOTE: Based on author's calculations using national data on employment and establishments for the childcare industry from the Statistics of United States Businesses, limited to NAICS 6244. Population statistics are compiled from the Surveillance, Epidemiology, and End Results Program.

Figure IV. Childcare Firm Characteristics

Panel A. Percent of Firms by Size

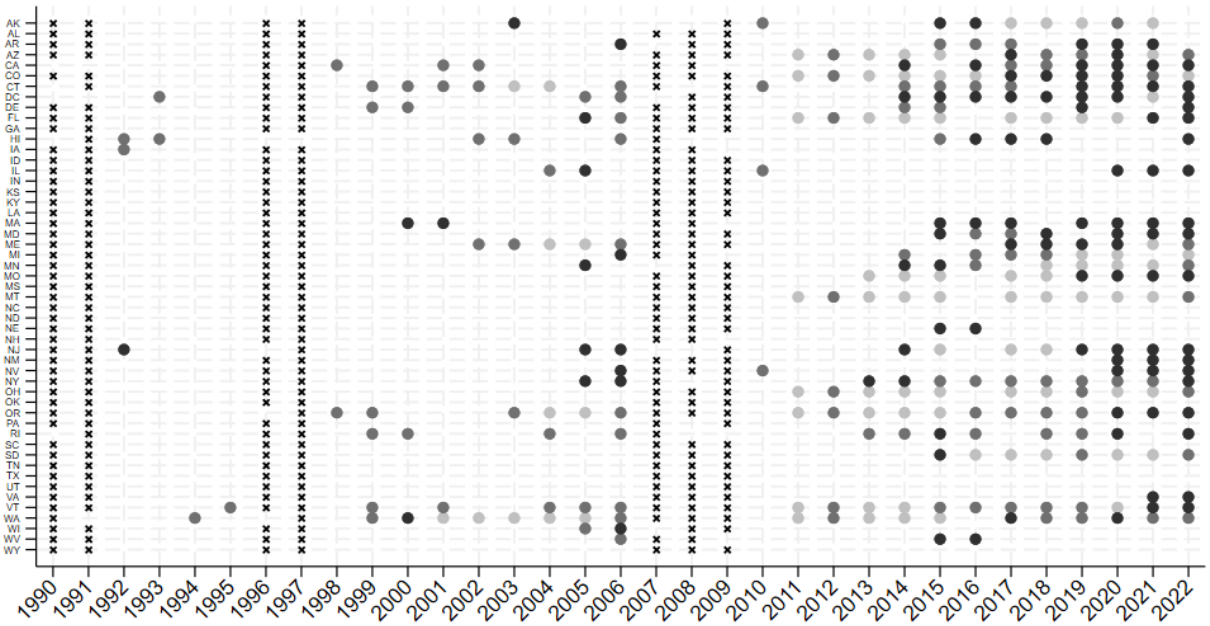


Panel B. Average Payroll per Employee by Firm Size



NOTE: Based on author's calculations using national data from the Statistics of United States Businesses. Figures in Panel B are in 2019 U.S. dollars.

Figure V. States Affected by Minimum Wage Policy Changes Over Time



NOTE: Based on author's calculations using minimum wage data from Vaghul & Zipperer (2016). X represents federal minimum wage changes, light gray circles represent a state minimum wage change between $0.01 and $0.25, medium gray circle represents a state minimum wage change between $0.26 and $0.75, and dark gray circle represents a state minimum change of over $0.76. If a state's minimum wage was already higher than the new federal minimum wage, they are considered unaffected.